26 Black Holes I: Approaching the Schwarzschild Radius r_s

Recall that the Schwarzschild metric, given in standard coordinates by

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \quad , \quad f(r) = 1 - \frac{2m}{r} \quad , \tag{26.1}$$

is the unique spherically symmetric vacuum solution of the Einstein equations. Associated with this solution there is a characteristic length scale, the *Schwarzschild radius* $r = r_s \equiv 2m$. More generally, this is the characteristic lenght scale associated to an object of mass M via the formula

$$r_s = 2m = \frac{2G_N M}{c^2} \quad . \tag{26.2}$$

In our previous discussions of this solution and its properties, in sections 24 and 25, we had considered standard astrophysical objects ("stars") of a size larger (in practice much larger) than their Schwarzschild radius, $r_0 > r_s = 2m$. As a consequence we did not have to address the question what happens to the metric or the geometry as $r \to r_s$ (which was in the deep interior of the star, not desribed by the Schwarzshild metric). We will now contemplate the existence of objects with radius $r_0 < r_s$.

26.1 Preliminary Considerations

Thus we now need to understand the behaviour of the Schwarzschild geometry / gravitational field as one approaches or crosses $r = r_s$. At that radius something special appears to happen to the metric, even though it is not clear what precisely happens there. On the one hand, we noted that the effective potential $V_{eff}(r)$ is perfectly well behaved at r_s. On the other hand, some components of the metric evidently become singular (zero or infinite) there.

Whether this indicates a true singularity of the geometry or the gravitational field (such as infinite tidal forces) or is perhaps simply due to an unfortunate choice of coordinates is usually not something that can just be decided by superficial inspection but requires a more detailed investigation.

Just to illustrate this point, it is a familiar fact that the (apparent) degeneracy of the Euclidean metric on $\mathbb{R}^2$ in polar coordinates,

$$ds^2 = dx^2 + dy^2 = dr^2 + r^2 d\phi^2 \quad , \tag{26.3}$$

at $r = 0$ is simply a coordinate singularity. Likewise, the fact that $g_{\phi\phi} \to \infty$ as $r \to \infty$ is not an indication of a singular behaviour of the metric there. By changing variables, $r = 1/u$, one finds that the line element takes the form

$$ds^2 = u^{-4}du^2 + u^{-2}d\phi^2 \quad . \tag{26.4}$$

In this case, both the non-zero components of the metric diverge as $u \to 0$, but again we know that this is just due to an unfortunate choice of coordinates. However, if one had just been given the above metric in the coordinates (u, ϕ) without an explanation how it was obtained, one might not have realised immediately that this is just the Euclidean metric in disguise. In fact, one can compare and contrast this e.g. with the metric

$$ds^2 = du^2 + u^{-2} d\phi^2 \quad , \tag{26.5}$$

which may look somewhat less singular than the previous metric but is actually genuinely singular as $u \to 0$. This can be verified by calculating its curvature following e.g. the examples given in section 11.3: then one finds that its Gauss curvature or Ricci scalar is proportional to u^{-2} (while the Gauss curvature of the Euclidean metric on $\mathbb{R}^2$ is of course identically zero, regardless of which coordinates one uses).

Thus in order to decide what happens or does not happen as $r \to r_s$, we need to take a closer look at the metric. We will do this in several steps, which you can think of as the various stages of an expedition, from the familiar region at $r \gg r_s$, with a map provided by the Schwarzschild coordinates (t, r, θ, ϕ), into potentially dangerous new and uncharted territory:

- First we will consider observers that don't quite dare to cross r_s and who try to remain static at a fixed value of r close to r_s.
- Then we will consider observers that fall freely (and radially) in this geometry, and describe their voyage both from their point of view and from that of a distant static observer (which will turn out to be quite different).
- Then we will study the geometry of the Schwarzschild metric near $r = r_s$, and show that the geometry is completely non-singular (and is in fact closely related to the geometry of the Rindler metric for Minkowski space-time discussed in section 1.3).
- These considerations will indicate that (and explain why) the usual Schwarzschild coordinates (specifically the time-coordinate t) are inadequate for describing the physics across the radius r_s.
- Encouraged by this, in section 27 we then begin to explore the region near and beyond $r = r_s$. To that end we first introduce coordinates that are adapted to infalling observers. This is an illuminating exercise which is interesting in its own right and which will already tell us something about the hidden geometry behind r_s.
- In order to learn more about this region, we next study the behaviour of lightcones and lightrays in this geometry. These considerations will then lead us to the introduction of corresponding adapted coordinates in which the Schwarzschild

metric is non-singular for all $0 < r < \infty$, and then we will be in a position to understand what actually happens (and what characterises) $r = r_s$.

26.2 Static Observers

Some insight into the Schwarzschild geometry, and the difference between Newtonian gravity and general relativistic gravity, is provided by looking at static observers, i.e. observers hovering at fixed values of (r, θ, ϕ). Thus their 4-velocity $u^\alpha = \dot{x}^\alpha$ has the form $u^\alpha = (u^0, 0, 0, 0)$ with $u^0 > 0$. The normalisation $u^\alpha u_\alpha = -1$ then implies

$$u^\alpha = \Big(f(r)^{-1/2}, 0, 0, 0\Big) = \left(\frac{1}{\sqrt{1-2m/r}}, 0, 0, 0\right) \quad . \tag{26.6}$$

Note that, as in (3.125), we could have written this more invariantly as

$$u^\alpha = V^\alpha / \mathbb{V} \tag{26.7}$$

where $V = \partial_t$ is the timelike Killing vector and $\mathbb{V}$ its norm (3.124), but for present (pragmatic and calculational) purposes the explicit coordinate expression is more useful.

The worldline of a static observer is clearly not a geodesic (that would be the worldline of an observer freely falling in the gravitational field), and we can calculate its covariant acceleration (5.98)

$$a^\alpha = D_\tau u^\alpha = u^\beta \nabla_\beta u^\alpha = (d/d\tau) u^\alpha + \Gamma^\alpha_{\beta\gamma} u^\beta u^\gamma \quad . \tag{26.8}$$

Noting that u^α is time-independent, because the observer is at fixed r by hypothesis, one finds

$$a^\alpha = \Gamma^\alpha_{00} (u^0)^2 = \Gamma^\alpha_{00} (1 - 2m/r)^{-1} \quad . \tag{26.9}$$

Thus only a^r is non-zero, and one finds

$$a^\alpha = (0, m/r^2, 0, 0) \quad . \tag{26.10}$$

This looks nicely Newtonian, with a force in the radial direction designed to precisely cancel the gravitational attraction. However, this is a bit misleading since this is still a coordinate dependent statement. A coordinate-invariant quantity (scalar) is the norm of the acceleration,

$$\mathsf{a}(r) \equiv \Big(g_{\alpha\beta} a^\alpha a^\beta\Big)^{1/2} = \frac{m}{r^2} \left(1 - \frac{2m}{r}\right)^{-1/2} \quad . \tag{26.11}$$

While this approaches the Newtonian value as $r \to \infty$, it diverges as $r \to 2m$, indicating that static observers will find it harder and harder to remain static close to $r = 2m$.

REMARKS:

1. At the same time, the discrepancy (gravitational time delay) between the static observer's proper time and the coordinate time (proper time of a static observer at $r \to \infty$) becomes more and more pronounced,

$$\Delta\tau = (1 - \frac{2m}{r})^{1/2}\Delta t \ , \tag{26.12}$$

and diverges as $r \to 2m$. This leads to an infinite gravitational redshift (see section 26.5 below). It is also the first (but all by itself not conclusive) indication, that the time coordinate t may not be appropriate for understanding the physics at and beyond $r = 2m$.

2. While $\mathsf{a}(r)$ diverges as $r \to 2m$, we will see later (section 27.10) that the finite quantity $\lim_{r\to 2m} f(r)^{1/2}\mathsf{a}(r) = 1/4m$ can be regarded as a measure of the strength of the gravitational field at $r = 2m$ (*surface gravity*).

3. One can also think of $\mathsf{a}(r) = \mathsf{a}^r(r)$ as the radial component of the acceleration with respect to an orthonormal frame (see section 4.8) at that point: a radial unit vector is

$$e_r = (1 - 2m/r)^{1/2}\partial_r : \quad g_{\alpha\beta}e_r^\alpha e_r^\beta = 1 \ , \tag{26.13}$$

and the acceleration vector can be written as

$$a = a^r\partial_r = a^r(1 - 2m/r)^{-1/2}((1 - 2m/r)^{1/2}\partial_r) \equiv \mathsf{a}^r e_r \ . \tag{26.14}$$

4. The covariant components of the acceleration can be written as the gradient

$$a_\alpha = \partial_\alpha(\tfrac{1}{2}\ln f(r)) \equiv \partial_\alpha\Phi(r) \ , \tag{26.15}$$

so that here (and in the static spherically symmetric case in general) $\Phi(r)$ can be regarded as the general relativistic analogue of the Newtonian potential $\phi(r)$, to which it reduces in the Newtonian limit,

$$\Phi(r) = \tfrac{1}{2}\ln f(r) \approx \tfrac{1}{2}\ln(1 + 2\phi(r)) = \phi(r) + \dots \tag{26.16}$$

5. Using the covariant characterisation (26.7) of the 4-velocity of a static observer, this result can be derived and generalised as follows. First of all the acceleration of u^α is

$$a_\beta = (V^\alpha/\mathbb{V})\nabla_\alpha(V_\beta/\mathbb{V}) = (V^\alpha/\mathbb{V}^2)\nabla_\alpha V_\beta \ , \tag{26.17}$$

because $\mathbb{V}$ is constant along V (9.61). Using the Killing condition this can be written as

$$a_\beta = -(V^\alpha/\mathbb{V}^2)\nabla_\beta V_\alpha = \tfrac{1}{2}\nabla_\beta\log(-V_\alpha V^\alpha) \ , \tag{26.18}$$

which provides the appropriate generalisation of (26.15), suggesting that a suitable general relativistic analogue of the scalar Newtonian potential in space-times admitting a timelike Killing vector is provided by the logarithm of the redshift factor $\mathbb{V}$,

$$\Phi = \tfrac{1}{2}\log(-V_\alpha V^\alpha) = \log\mathbb{V} \ . \tag{26.19}$$

26.3 Vertical Free Fall

We will now consider an object with $r_0 < r_s$ and an observer who is freely falling vertically (radially) towards such an object. "Vertical" means that $\dot{\phi} = 0$, and therefore there is no angular momentum, $L = 0$. Hence the effective potential equation (25.31) becomes

$$E^2 - 1 = \dot{r}^2 - \frac{2m}{r} \quad , \tag{26.20}$$

where the conserved energy E (per unit mass) is given by

$$E = (1 - \frac{2m}{r})\dot{t} \quad . \tag{26.21}$$

In particular, if r_i is the point at which the particle (observer) A was initially at rest,

$$\frac{dr}{d\tau}|_{r=r_i} = 0 \quad , \tag{26.22}$$

we have the relation

$$E^2 = 1 - \frac{2m}{r_i} = f(r_i) \tag{26.23}$$

between the constant of motion E and the initial condition r_i. In particular, $E = 1$ for an observer following a trajectory of an object that would have initially been at rest at infinity. In that case, (26.20) is readily integrated to give

$$r(\tau) \sim (\tau_0 - \tau)^{2/3} \tag{26.24}$$

so that the observer reaches $r = 0$ at $\tau = \tau_0$.

For $r_i < \infty$, we obtain

$$\dot{r}^2 = \frac{2m}{r} - \frac{2m}{r_i} = f(r_i) - f(r) \tag{26.25}$$

and, upon differentiation (for any E)

$$\ddot{r} + \frac{m}{r^2} = 0 \quad . \tag{26.26}$$

This is just like the Newtonian equation (which should not come as a surprise as V_{eff} coincides with the Newtonian potential for zero angular momentum $L = 0$), apart from the fact that r is *not* radial distance and the familiar $\tau \neq t$. Nevertheless, calculation of the time τ along the path proceeds exactly as in the Newtonian theory. For the proper time required to reach the point with coordinate value $r = r_f$ we obtain

$$\tau = -(2m)^{-1/2} \int_{r_i}^{r_f} dr \, \left(\frac{r_i r}{r_i - r} \right)^{1/2} \quad . \tag{26.27}$$

Since this is just the Newtonian integral, we know, even without calculating it, that it is finite as $r_f \to r_s$ (and even as $r_f \to 0$). This integral can also be calculated in closed form, e.g. via the change of variables

$$r(\eta) = \tfrac{1}{2} r_i \, (1 + \cos \eta) \quad \text{with} \quad \eta_i = 0 \leq \eta \leq \eta_f \leq \pi \quad , \tag{26.28}$$

leading (after some convenient cancellations) to

$$\tau = \left(\frac{r_i}{2m}\right)^{1/2} \int_0^{\eta_f} d\eta\, r(\eta) = \left(\frac{r_i^3}{8m}\right)^{1/2} (\eta_f + \sin\eta_f) \ . \tag{26.29}$$

In particular, this is finite as $r_f \to 2m$ and our freely falling observer can reach and cross the Schwarzschild radius r_s in finite proper time.

Coordinate time, on the other hand, becomes infinite at $r_f = 2m$. This can be anticipated from the relation (26.21) between t and τ for a freely falling observer, which diverges as $r \to 2m$. We will address this in a more quantitative way in section 26.4 below.

26.4 Vertical Free Fall as seen by a Distant Observer

We will now investigate how the above situation presents itself to a distant observer hovering at a fixed radial distance r_∞. He will observe the trajectory of the freely falling observer as a function of his proper time τ_∞. Up to a constant factor $(1 - 2m/r_\infty)^{1/2}$, this is the same as coordinate time t, and we will lose nothing by expressing r as a function of t rather than as a function of τ_∞.

From (26.20),

$$\dot{r}^2 + (1 - \frac{2m}{r}) = E^2 \ , \tag{26.30}$$

which expresses r as a function of the freely falling observer's proper time τ, and the definition of E,

$$E = \dot{t}(1 - \frac{2m}{r}) \ , \tag{26.31}$$

which relates τ to the coordinate time t, one finds an equation for r as a function of t,

$$\frac{dr}{dt} = -E^{-1}(1 - \frac{2m}{r})(E^2 - (1 - \frac{2m}{r}))^{1/2} \tag{26.32}$$

(the minus sign has been chosen because r decreases as t increases). At large distances, the difference between t and τ, and hence between $r = r(t)$ and $r = r(\tau)$ is small and not particularly interesting in the present context. We want to analyse the behaviour of the solution of this equation as the freely falling observer approaches the Schwarzschild radius, $r \to 2m$. In that region, we can approximate

$$1 - \frac{2m}{r} = \frac{r - 2m}{r} \approx \frac{r - 2m}{2m} \ , \tag{26.33}$$

and we get

$$\begin{aligned} \frac{dr}{dt} &= -E^{-1}(\frac{r - 2m}{r})(E^2 - \frac{r - 2m}{r})^{1/2} \\ &\approx -E^{-1}(\frac{r - 2m}{2m})(E^2)^{1/2} = -(\frac{r - 2m}{2m}) \ . \end{aligned} \tag{26.34}$$

We can write this equation as

$$\frac{d}{dt}(r-2m) = -\frac{1}{2m}(r-2m) \quad , \tag{26.35}$$

which obviously has the solution

$$(r-2m)(t) \propto \mathrm{e}^{-t/2m} \quad . \tag{26.36}$$

This shows that, from the point of view of the observer at infinity, the freely falling observer reaches $r = 2m$ only as $t \to \infty$. In particular, the distant observer will never actually see the infalling observer cross the Schwarzschild radius.

This is clearly an indication that there is something wrong with the time coordinate t which runs too fast as one approaches the Schwarzschild radius. We can also see this by looking at the coordinate velocity $v = dr/dt$ as a function of r. Let us choose $r_i = \infty$ for simplicity - other choices will not change our conclusions as we are interested in the behaviour of $v(r)$ as $r \to r_s$. Then $E^2 = 1$ and from (26.32) we find (now dropping the minus sign)

$$v(r) = (2m)^{1/2}\frac{r-2m}{r^{3/2}} \tag{26.37}$$

As a function of r, $v(r)$ reaches a maximum at the critical radius $r_c = 6m = 3r_s$,

$$\frac{d}{dr}v(r) = 0 \quad \Rightarrow \quad r = r_c = 6m \quad , \tag{26.38}$$

where the velocity is (restoring the speed of light c)

$$v(r_c) = \frac{2c}{3\sqrt{3}} \quad . \tag{26.39}$$

The fact that this radius agrees with the ISCO (innermost stable circular orbit) (25.73) mentioned in section 25.6 should (presumably) be regarded as a coincidence.

Beyond that point, i.e. for $r < r_c$, $v(r)$ decreases again and clearly goes to zero as $r \to 2m$. The fact that the coordinate velocity goes to zero is simply another manifestation of the fact that coordinate time goes to infinity, and that the Schwarzschild coordinates are simply not suitable for describing the physics at or beyond the Schwarzschild radius because the time coordinate one has chosen is running too fast. This is the crucial insight that will later on allow us to construct "better" coordinates, which are also valid for $r < r_s$.

Remarks:

1. Sometimes (actually all too frequently) the fact that the coordinate velocity is decreasing as one approaches $r \to 2m$ is claimed to provide evidence for some "repulsive" nature of the gravitational field as one approaches $r = 2m$. While it is understandable that such things caused some confusion in the early days of general relativity, this is of course utter nonsense for which there can be no excuse today.

2. As an aside, if one repeats the above "calculation" for arbitrary E, from (26.32) one finds that the critical radius is

$$r_c = \frac{2m}{1 - 2E^2/3} \quad , \tag{26.40}$$

with the E-dependent maximal velocity

$$v(r_c) = \frac{2c}{3\sqrt{3}} E^2 \quad . \tag{26.41}$$

This reproduces the above result for $E = 1$, shows that $r_c \to 2m$ for $E \to 0$, and moreover curiously shows that there is a maximal E, $E_{max} = \sqrt{3/2}$, for which such a critical point of the coordinate velocity will occur, at $r_c \to \infty$, with maximal velocity

$$E \to E_{max} = \sqrt{3/2} \quad \Rightarrow \quad r_c \to \infty \quad \text{and} \quad v(r_c) \to v_{max} = c/\sqrt{3} \ . \tag{26.42}$$

Thus for larger values of E, the coordinate velocity will monotonically decrease from its initial value

$$v(r \to \infty) = E^{-1}(E^2 - 1)^{1/2} \tag{26.43}$$

to $v(r) = 0$ as $r \to 2m$.

[And if anybody has an intuitive explanation for these facts, please let me know.]

26.5 Gravitational Redshift in the Schwarzschild Geometry

One dramatic aspect of what is happening at (or, better, near) the Schwarzschild radius for very (very!) compact objects with $r_s > r_0$ is the following. Recall the formula (3.108) for the gravitational redshift, which gave us the ratio between the frequency of light ν_e emitted at the radius r_e and the frequency ν_∞ received at the radius $r_\infty > r_e$ for static observers in a static spherically symmetric gravitational field.

The result, which is in particular also valid for the Schwarzschild metric, was

$$\frac{\nu_\infty}{\nu_e} = \frac{(-g_{00}(r_e))^{1/2}}{(-g_{00}(r_\infty))^{1/2}} \quad . \tag{26.44}$$

In the case of the Schwarzschild metric, this is

$$\frac{\nu_\infty}{\nu_e} = \frac{(1 - 2m/r_e)^{1/2}}{(1 - 2m/r_\infty)^{1/2}} \quad , \tag{26.45}$$

in accordance with the relation (26.12) between the coordinate time and the proper time of a static observer. As $r_e \to r_s$, one clearly finds

$$\frac{\nu_\infty}{\nu_e} \to 0 \quad . \tag{26.46}$$

Expressed in terms of the gravitational redshift factor z,

$$1 + z = \frac{\nu_e}{\nu_\infty} \tag{26.47}$$

this means that there is an *infinite gravitational redshift* as $r_e \to r_s$,

$$r_e \to r_s \quad \Rightarrow \quad z \to \infty \ . \tag{26.48}$$

This is for static observers. For a freely falling observer whose position is described by $r_e = r(\tau)$ or $r(t)$ and the receiver the static observer at $r_\infty \gg r_s$ we cannot directly apply the above formula (because the emitter is not static but freely falling), and the actual redshift of the emitted lightray will be larger because of an additional Doppler effect due to the observer's proper motion away from the receiver.

If (for the time being) we ignore this additional Doppler contribution, and just use the above formula (this will thus underestimate the actual redshift) in conjunction with the late-time behaviour for $r = r(t)$ derived in (26.36), we can obtain the late-time behaviour of the redshift factor $z = z(t)$ as a function of the distant observer's proper time t, namely

$$1 + z \propto (r - 2m)(t)^{-1/2} \propto \mathrm{e}^{t/4m} \ . \tag{26.49}$$

Taking into account the proper motion of the freely falling observer one finds (we will derive this below) that, essentially due to the difference between $u^0 = \dot{t} = f^{-1/2}$ for a static observer and $u^0 = \dot{t} = E/f$ for a geodesic observer, as $r \to 2m$ the redshift is enhanced by a further factor $f^{-1/2}$, so that all in all one has $(1 + z) \propto (r - 2m)(t)^{-1}$ instead of (26.49). See (26.67) below.

Either way for the distant observer at late times there is an exponentially growing redshift and the distant observer will never actually see the unfortunate emitter crossing the Schwarzschild radius: he/she will see the freely falling observer's signals becoming dimmer and dimmer and arriving at greater and greater intervals, and the freely falling observer will completely disappear from the distant observer's sight as $r_e \to r_s$. Note that the time-scale t_z for this exponential redshift at late times is set by $t_z \sim m/c$, which is of the order of

$$t_z \sim 10^{-5}\mathsf{s} \left(\frac{M}{M_{\mathsf{sun}}} \right) \ , \tag{26.50}$$

so that this is pretty much instantaneous for an object the mass of an ordinary star. We will come back to this estimate later on when (briefly) talking about gravitational collapse in section 29.3.

In order to take into account the proper freely falling motion of the emitter, one can use the method described in the second derivation of the gravitational redshift given in section 3.5, based on the equation (3.117),

$$\omega = -u^\alpha k_\alpha \ , \tag{26.51}$$

for the frequency ω of a lightray with wave vector k^α as seen (or emitted) by an arbitrary (not necessarily static) observer with 4-velocity u^α.

We describe lightrays by null curves $x^\alpha = x^\alpha(\lambda)$ with wave vector

$$k^\alpha = \frac{d}{d\lambda} x^\alpha(\lambda) \equiv x'^\alpha \tag{26.52}$$

(for some affine parameter λ) and the motion of timelike observers (as usual) by $x^\alpha = x^\alpha(\tau)$ with 4-velocity

$$u^\alpha = \frac{d}{d\tau} x^\alpha(\tau) \equiv \dot{x}^\alpha \quad . \tag{26.53}$$

Let us now apply this formula to the case at hand (Schwarzschild geometry and radial motion). The relevant wave and velocity vectors are the following:

- Radial Lightrays

  For lightrays, there is a conserved energy

  $$e = f(r) t' \tag{26.54}$$

  and for radial lightrays the effective potential is zero, so that the radial equation of motion for r is simply

  $$(r')^2 = e^2 \quad \Rightarrow \quad r' = \pm e \tag{26.55}$$

  (for out/in-going lightrays respectively). Thus out/in-going radial lightrays have wave vector

  $$(k^\alpha)_{out/in} = (t', r', 0, 0) = (e/f, \pm e, 0, 0) : \quad g_{\alpha\beta} k^\alpha k^\beta = 0 \quad . \tag{26.56}$$

- Static Observers

  For static observers one has (26.6)

  $$(u^\alpha)_s = (f^{-1/2}, 0, 0, 0) : \quad g_{\alpha\beta} u^\alpha u^\beta = -1 \quad . \tag{26.57}$$

- Radial Freely Falling Observers

  For geodesic (freely falling) obervers, there is a conserved energy

  $$E = f(r) \dot{t} \tag{26.58}$$

  and for radial free fall the effective potential is just the Newtonian potential, so that the radial equation of motion is simply (26.20)

  $$\dot{r}^2 = E^2 - f \quad \Rightarrow \quad \dot{r} = -(E^2 - f)^{1/2} \tag{26.59}$$

  (for infalling observers). Thus their 4-velocity is

  $$(u^\alpha)_{ff} = (\dot{t}, \dot{r}, 0, 0) = (E/f, -(E^2 - f)^{1/2}, 0, 0) : \quad g_{\alpha\beta} u^\alpha u^\beta = -1 \quad . \tag{26.60}$$

From these and (26.51) we can now read off the following results:

1. Static Observers and In/Outgoing Lightrays

   The frequency of an in/out-going lightray as seen by a static observer at the radius r is

   $$\omega_{s,in/out}(r) = -g_{\alpha\beta}(u^\alpha)_s(k^\beta)_{in/out} = f(r)f(r)^{-1/2}(e/f(r)) = ef(r)^{-1/2} \ . \tag{26.61}$$

   Note that this only depends on the time component of k^α and is thus the same for in- and outgoing lightrays. In particular,

   - this identifies the energy e as the frequency as seen by an observer "at infinity",
   $$\omega_\infty \equiv \lim_{r\to\infty} \omega_{s,in/out} = e \ , \tag{26.62}$$
   - and reproduces the standard result that the redshift of light emitted by a static observer at radius r relative to a distant static observer scales like $f(r)^{-1/2}$,
   $$\frac{\omega_\infty}{\omega_{s,in/out}} = f(r)^{1/2} \ . \tag{26.63}$$

2. Freely Falling Observers and In- versus Outgoing Lightrays

   For a freely falling observer instead, one has

   $$\omega_{ff,in/out} = -g_{\alpha\beta}(u^\alpha)_{ff}k^\beta_{in/out} = f(E/f)(e/f) \mp f^{-1}(E^2 - f)^{1/2}e \tag{26.64}$$

   and thus (restoring the r-dependence)

   $$\omega_{ff,in/out}(r) = \frac{e}{f(r)}(E \mp (E^2 - f(r))^{1/2}) \ . \tag{26.65}$$

   Note that, roughly speaking, and as anticipated above, compared to the result for a static emitter or observer, the redshift/blueshift is enhanced by another factor $f(r)^{\pm 1/2}$ due to a Doppler-like effect caused by the motion of the observer or emitter.

   This effect is of course particularly pronounced if either emitter or observer approaches $r = 2m$. More specifically, we can consider the following two situations:

   (a) Outgoing Lightrays

   Taking the limit $f(r) \to 0$, one finds that

   $$f(r) \to 0 \quad \Rightarrow \quad \omega_{ff,out}(r) \approx \frac{2eE}{f(r)} \quad \Rightarrow \quad \frac{\omega_\infty}{\omega_{ff,out}(r)} \approx \frac{f(r)}{2E} \to 0 \ . \tag{26.66}$$

   Thus from the perspective of a static observer an outgoing lightray sent out by an infalling observer will undergo an infinite redshift proportional to

$(r-2m)^{-1}$ as the emitter approaches $r \to 2m$. In particular, here we are now allowed to insert the time dependence (26.36) and instead of (26.49) we will obtain

$$1+z \propto (r-2m)(t)^{-1} \propto \mathrm{e}^{\,t/2m} \quad . \tag{26.67}$$

(b) Ingoing Lightrays

We can also consider the opposite situation where an ingoing lightray reaches an infalling observer (from above/behind) as $r \to 2m$. In that case, the lightray is detected by the infalling observer with a finite redshift,

$$f(r) \to 0 \quad \Rightarrow \quad \omega_{ff,in}(r) \approx \frac{e}{2E} \quad \Rightarrow \quad \frac{\omega_\infty}{\omega_{ff,in}} = 2E \quad . \tag{26.68}$$

In particular, for a freely falling observer with $E = 1$, i.e. an oberver who falls as if he/she had started off at rest at infinity, the frequency he/she measures is precisely 1/2 the frequency ω_∞.

This is perhaps one "practical" way for this infalling observer to determine when precisely he/she reaches $r = 2m$: just after the agreed upon almost-UV violet light with emitter frequency 790 THz (at the upper end of the visible spectrum) arrives as the almost-IR red light with frequency 405 THz (at the lower end of the visible spectrum).

26.6 Geometry near r_s and Minkowski Space in Rindler Coordinates

We have now seen in two different ways why the Schwarzschild coordinates are not suitable for exploring the physics in the region $r \leq 2m$: in these coordinates the metric becomes singular at $r = 2m$ and the coordinate time becomes infinite. On the other hand, we have seen no indication that the local physics, expressed in terms of covariant quantities like proper time or the geodesic equation, becomes singular as well. So we have good reasons to suspect that the singular behaviour we have found is really just an artefact of a bad choice of coordinates.

In fact, the situation regarding the Schwarzschild coordinates is quite similar to that of the Rindler coordinates for Minkowski space we discussed (way back) in section 1.3.

Recall from section 1.3 that

- in the coordinates

$$\xi^0(\eta,\rho) = \rho \sinh \eta \qquad \xi^1(\eta,\rho) = \rho \cosh \eta \quad . \tag{26.69}$$

the $(1+1)$-dimensional Minkowski metric takes the form (1.74)

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 \quad ; \tag{26.70}$$

- that the lines of constant ρ are hyperbolas and that these are the worldlines of observers with constant acceleration $1/\rho$;
- that these coordinates are adapted to observers with constant acceleration in the same way as inertial coordinates are adapted to static observers: they stay at fixed values of their spatial coordinate, and the coordinate time is a direct measure of their proper time;
- the null lines $\xi^0 = \pm\xi^1$ correspond to $\rho = 0$ (infinite acceleration) or $\eta = \pm\infty$, so that the Rindler coordinates cover only the first quadrant $\xi^1 > |\xi^0|$ of Minkowski space;
- inertial (geodesic, freely falling) observers could exit this region in finite proper time.

All this is of course quite reminiscent of the things that we have discovered so far about the Schwarzschild geometry, and this is in fact more than just a loose analogy: as we will see now, remarkably the Rindler metric (26.70) gives an accurate description of the geometry of the Schwarzschild metric close to the Schwarzschild radius.

To confirm this, let us temporarily introduce the variable $\tilde{r} = r - 2m$ measuring the coordinate distance from the critical radius $r = 2m$. In term of $\tilde{r}$ the (t, r)-part of the Schwarzschild metric reads

$$ds^2 = -\left(\frac{\tilde{r}}{\tilde{r}+2m}\right)dt^2 + \left(\frac{\tilde{r}+2m}{\tilde{r}}\right)d\tilde{r}^2 \quad . \tag{26.71}$$

Close to $r_s = 2m$, i.e. for small $\tilde{r} << 2m$, we can approximate

$$\frac{\tilde{r}}{\tilde{r}+2m} \approx \frac{\tilde{r}}{2m} \qquad\qquad \frac{\tilde{r}+2m}{\tilde{r}} \approx \frac{2m}{\tilde{r}} \quad , \tag{26.72}$$

so that the metric becomes

$$ds^2 = -\frac{\tilde{r}}{2m}dt^2 + \frac{2m}{\tilde{r}}d\tilde{r}^2 \quad . \tag{26.73}$$

Introducing the new radial variable ρ (proper radial distance from $r = r_s$) via

$$d\rho^2 = \frac{2m}{\tilde{r}}d\tilde{r}^2 \quad \Rightarrow \quad \rho = \sqrt{8m\tilde{r}} \quad , \tag{26.74}$$

one finds

$$ds^2 = -\frac{1}{16m^2}\rho^2 dt^2 + d\rho^2 \quad . \tag{26.75}$$

Finally a simple rescaling of t, $\eta = t/4m$, leads to

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 \quad , \tag{26.76}$$

which, remarkably, is identical to the Rindler metric (1.74). Keeping track of the transverse 2-sphere and using $r^2 \approx (2m)^2$ in the near-r_s approximation, the complete metric in this limit and in these coordinates reads

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2 + (2m)^2 d\Omega^2 \tag{26.77}$$

If we further restrict to just a small angular region on the sphere, we can approximate

$$d\theta^2 + \sin^2\theta d\phi^2 \approx d\theta^2 + \theta^2 d\phi^2 = (dx^2)^2 + (dx^3)^2 \tag{26.78}$$

which gives us the complete 4-dimensional Rindler metric.

In any case, we see that, up to the harmless redefinition $(r,t) \to (\rho,\eta)$ that we just performed, which is just a reparametrisation

$$\begin{aligned} \rho &= \rho(r) = \sqrt{8m(r-2m)} \\ \eta &= \eta(t) = t/4m \end{aligned} \tag{26.79}$$

Schwarzschild coordinates for the Schwarzschild geometry near $r = 2m$ are just like Rindler coordinates for Minkowski space. This leads to a much improved understanding of the Schwarzschild geometry in general and the observations that we made regarding static and freely falling observers in particular:

1. The first, and most crucial, thing we learn from this is that the singularity of the Schwarzschild metric at $r = 2m$ in the Schwarzschild coordinates (t,r) is, as anticipated, a mere coordinate singularity. Indeed, $r = 2m$ corresponds to $\tilde{r} = 0 \Leftrightarrow \rho = 0$, and we already know that what appears to be a possibly singular point of the Rindler metric at $\rho = 0$ is just a coordinate singularity which can be eliminated e.g. by passing to standard inertial Minkowski coordinates via (1.73).

   Specifically, from (26.79) one could introduce Minkowski-like coordinates (t_M, x_M) (say), via

   $$t_M = \sqrt{8m(r-2m)}\sinh t/4m \quad , \quad x_M = \sqrt{8m(r-2m)}\cosh t/4m \ , \tag{26.80}$$

   in terms of which the metric in the region $r \gtrsim 2m$ takes the standard Minkowskian form $-dt_M^2 + dx_M^2$. The reason for not pursuing this here is that we will actually be able to do much better than this in various ways in section 27, where we will construct a variety of coordinate systems which exhibit the non-singular nature of the geometry at $r = 2m$ but which are not restricted to only a tiny (strictly speaking infinitesimal) neighbourhood of $r = 2m$. In particular, the above coordinate transformation (26.80) can be understood as the $r \gtrsim 2m$ approximation of the coordinate transformation from Schwarzschild coordinates to *Kruskal coordinates* (to be introduced in section 27.7).

2. Moreover, we can now understand physically why the Schwarzschild coordinates break down at $r = 2m$: they are adapted to accelerating observers, in the present context the static observers in the Schwarzschild geometry at constant r, with proper time proportional to the Schwarzschild coordinate time t. Referring back to Figure 7 in section 1.3, these are the observers with constant ρ hyperbolic world lines.

   The problem is evidently that the required acceleration of these observers becomes infinite as $\rho \to 0 \Leftrightarrow r \to 2m$ (as we have calculated in section 26.2). That these observers appear to see a singular metric is then not the geometry's fault but can be attributed to a bad choice of observers whose perception of the geometry is distorted by their acceleration and their desperate attempt to stay at constant values of r even when they are very close to $r = 2m$.

3. The situation is quite different for the freely falling observers. Their worldlines look like the vertical line labelled "worldline of a static observer" in Figure 7, they cross the "horizon" in finite proper time, experiencing no strong acceleration or gravitational fields. As already noted in section 1.3, they evidently become invisible to observers in the Rindler quadrant (now "Schwarzschild patch") of the geometry, static outside observers noting an infinite gravitational redshift affecting the signals sent out by the freely falling observer.

4. Introducing coordinates adapted to freely falling observers (so that e.g. time is their proper time) would be tantamount to passing from Rindler coordinates to ordinary Minkowski coordinates, and we will consider that option in section 27.2 below (Painlevé-Gullstrand coordinates). These will provide us, as we will see, with a coordinate system that extends in a non-singular way across the Schwarzschild radius.

5. In order to elucidate the significance of the Schwarzschild radius, it will then turn out to be useful to base the construction of new coordinates not on the behaviour of timelike freely falling observers but on ingoing lightrays instead (which evidently also do not suffer from the problems of the static observers). These are the Eddington-Finkelstein coordinates to be discussed in section 27.4.

6. Finally, we can anticipate that upon introduction of suitable analogues of the Minkowski coordinates for the Rindler space-time, we may perhaps uncover not just one new region (quadrant) of space-time (the one lying to the "future" of $r = 2m$), but also counterparts of the other two quadrants of Minkowski space. This expectation will indeed be borne out.

In section 26.3 we had seen that observers following timelike radial ingoing geodesics reach $r = 2m$ at finite values of the affine parameter (proper time), and in section 26.4 that they do so at infinite values of the coordinate time t. The same is *mutatis mutandis* true for ingoing lightrays travelling along null geodesics, and the argument in this case is even simpler.

Indeed, since for radial lightrays the effective potential (25.43) is zero, radial lightrays are governed by the equation

$$\tfrac{1}{2}\dot{r}^2 = E_{eff} = \tfrac{1}{2}E^2 \quad \Rightarrow \quad \dot{r} = \pm E \ , \tag{26.81}$$

the lower sign corresponding to ingoing lightrays. The solution for ingoing lightrays is thus evidently

$$r(\lambda) = r(0) - E\lambda \tag{26.82}$$

so that from any finite initial position $r(0) > 2m$, the Schwarzschild radius $r = 2m$ is reached for the finite value

$$\lambda = \frac{r(0) - 2m}{E} \tag{26.83}$$

of the affine parameter.

Note, by the way, that (26.82) or $\ddot{r} = 0$ shows that r and λ are related by an affine transformation so that one can equally well use r as the affine parameter along ingoing null geodesics, making it even more manifest that $r = 2m$ arises at a finite value of that affine parameter. To reduce clutter, let us choose $r(0) = 2m$ so that $r = 2m$ is reached at $\lambda = 0$. It then follows from $f(r)\dot{t} = E$, i.e.

$$\dot{t} = E/f(r) = Er/(r - 2m) = E + 2mE/(r - 2m) \ , \tag{26.84}$$

that

$$\dot{t}(\lambda) = E + 2mE/(-E\lambda) = E - 2m/\lambda \quad \Rightarrow \quad t(\lambda) = E\lambda - 2m\log(-\lambda) \ . \tag{26.85}$$

This makes it manifest that

$$r \to 2m \quad \Rightarrow \quad \lambda \to 0_- \quad \Rightarrow \quad t \to +\infty \tag{26.86}$$

also for null geodesics. Thus t is evidently not a good coordinate to describe physics at (or beyond) $r = 2m$.

To improve our understanding of the Schwarzschild geometry, it is important to study its *causal structure*, i.e. the lightcones. Radial null curves satisfy

$$(1 - 2m/r)dt^2 = (1 - 2m/r)^{-1}dr^2 \ . \tag{26.87}$$

Thus

$$\frac{dt}{dr} = \pm(1 - 2m/r)^{-1} \ . \tag{26.88}$$

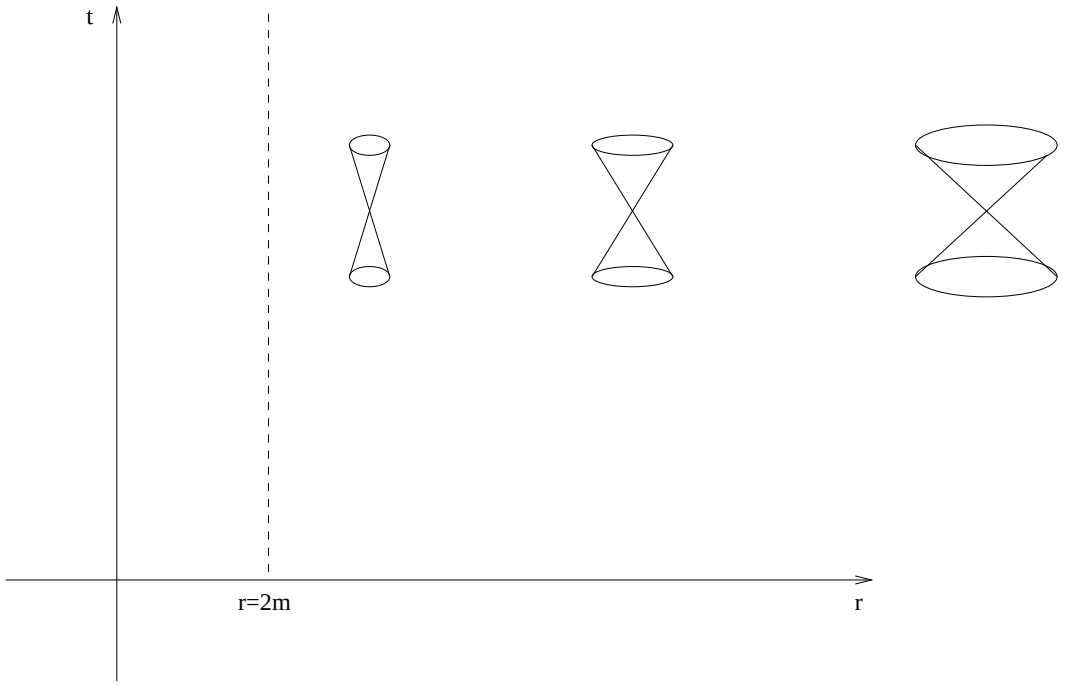


Figure 16: Causal structure of the Schwarzschild geometry in the Schwarzschild coordinates (r, t). As one approaches $r = 2m$, the lightcones become narrower and narrower and eventually fold up completely.

Recall that, as noted above, this equation for $t = t(r)$ can be equivalently regarded as an equation for $t = t(\lambda)$ as a function of the affine parameter.

In the (t, r)-diagram of Figure 16, dt/dr represents the slope of the lightcones at a given value of r. Now, as $r \to 2m$, one has

$$\frac{dt}{dr} \stackrel{r\to 2m}{\longrightarrow} \pm\infty \ , \tag{26.89}$$

so the light cones 'close up' as one approaches the Schwarzschild radius. This is the same statement as before regarding the fact that the coordinate velocity goes to zero at $r = 2m$, but this time for null rather than timelike geodesics.

As our first step towards introducing coordinates that are more suitable for describing the region around r_s, let us write the Schwarzschild metric in the form

$$ds^2 = (1 - 2m/r)(-dt^2 + (1 - 2m/r)^{-2} dr^2) + r^2 d\Omega^2 \ . \tag{26.90}$$

We see that it is convenient to introduce a new radial coordinate r^* via

$$dr^* = (1 - 2m/r)^{-1} dr = \frac{r}{r - 2m} dr = (1 + \frac{2m}{r - 2m}) dr \ . \tag{26.91}$$

The solution to this equation is (up to an arbitrary finite constant)

$$r^* = r + 2m \log(r/2m - 1) \ . \tag{26.92}$$

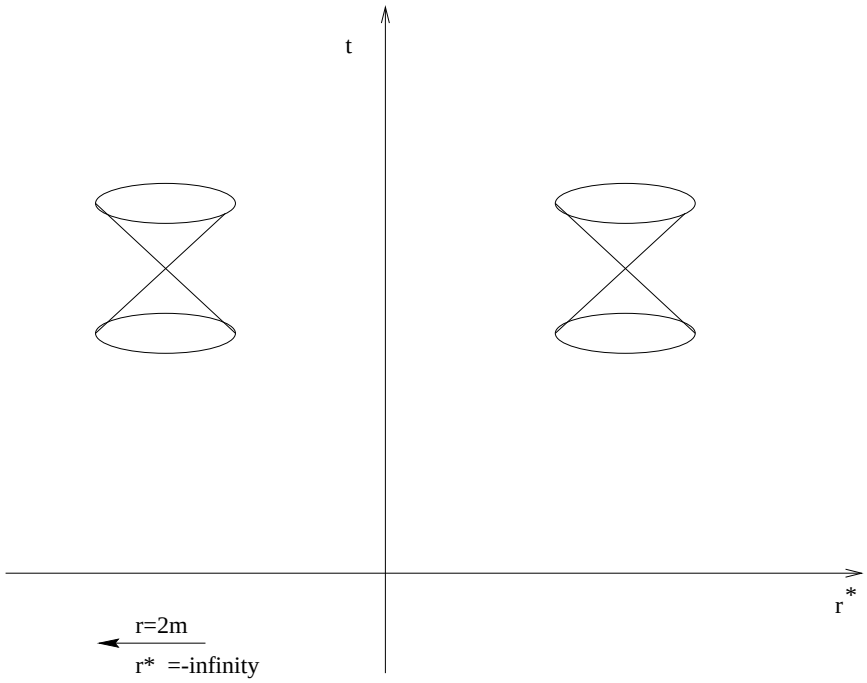


Figure 17: Causal structure of the Schwarzschild geometry in the tortoise coordinates (r^*, t). The lightcones look like the lightcones in Minkowski space and no longer fold up as $r \to 2m$ (which now sits at $r^* = -\infty$).

This new radial coordinate r^*, known as the *Regge - Wheeler radial coordinate* or *tortoise coordinate*, also provides us with the solution

$$t = \pm r^* + C_\pm \tag{26.93}$$

(with $C_\pm$ constants of integration) to the equation (26.88) describing the radial lightcones:

- lines of constant $C_+ = t - r^*$ describe *outgoing* lightrays;
- lines of constant $C_- = t + r^*$ describe *ingoing* lightrays.

In terms of r^* the metric simply reads

$$ds^2 = (1 - 2m/r)(-dt^2 + dr^{*2}) + r^2 d\Omega^2 \ , \tag{26.94}$$

where r is to be thought of as a function of r^*.

Now the lightcones, defined by

$$dt^2 = dr^{*2} \ , \tag{26.95}$$

do not seem to fold up as the lightcones have the constant slope $dt/dr^* = \pm 1$ (see Figure 17), and there is no singularity at $r = 2m$. However, r^* is still only defined for

$r > 2m$ and the surface $r = 2m$ has been pushed infinitely far away ($r = 2m$ is now at $r^* = -\infty$). Moreover, even though non-singular, the metric components g_{tt} and $g_{r^*r^*}$ (as well as $\sqrt{g}$) vanish at $r = 2m$.

Thus the tortoise coordinate has so far not really allowed us to make dramatic progress in our exploration of the region near or behind r_s, but we will substantially improve this situation in section 27.

26.8 Massless Klein-Gordon Scalar Field in the Schwarzschild Geometry

As an aside, and to conclude this section, I just want to point out that the tortoise coordinate r^* and the corresponding retarded and advanced time-coordinates $u, v = t \mp r^*$ (which we will reencounter in section 27.4 as part of the Eddington Finkelstein coordinates) are not only useful for clarifying the causal structure of the Schwarzschild geometry, but also for the analysis of the propagation of scalar (and other) fields. This is principally due to the fact that in these coordinates the (t, r)-part of the metric is conformally flat - see (26.94) or (27.141) below. Combined with the observation of section 7.7 that the massless Klein-Gordon action is conformally invariant in $(1 + 1)$-dimensions, this leads to a canonical form for the (3+1) wave operator in the (t, r)-sector (and the (θ, ϕ)-sector is standard anyway).

Specifically, we start with the action for a massless scalar field ϕ in the metric (26.94),

$$\begin{aligned} S[\phi] &\sim \int \sqrt{g} d^4x \; g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi \\ &= \int dt \; dr^* \; d\Omega \; r^2 f(r) \left[f(r)^{-1} (-(\partial_t \phi)^2 + (\partial_{r^*} \phi)^2) - r^{-2} \phi \; \Delta_{S^2} \phi \right] \\ &= \int dt \; dr^* \; d\Omega \; \left[-(r\partial_t \phi)^2 + (r\partial_{r^*} \phi)^2) - f(r) \phi \; \Delta_{S^2} \phi \right] \quad , \end{aligned} \tag{26.96}$$

where $f(r) = 1 - 2m/r$, $d\Omega = \sin\theta d\theta \; d\phi$ denotes the solid angle on the 2-sphere, and Δ_{S^2} is the Laplace operator on the 2-sphere. Separating variables according to

$$\phi(x) = r^{-1} \sum_{\ell,m} \psi_{\ell m}(t, r^*) Y_{\ell m}(\theta, \phi) \quad , \tag{26.97}$$

using

$$\Delta_{S^2} Y_{\ell m} = -\ell(\ell + 1) Y_{\ell m} \quad , \tag{26.98}$$

and using

$$\partial_{r^*} r = f(r) \quad \Rightarrow \quad r\partial_{r^*}(\psi_{\ell m}/r) = \partial_{r^*} \psi_{\ell m} - r^{-1} f(r) \psi_{\ell m} \quad , \tag{26.99}$$

one finds for $\psi = \psi_{\ell m}$ the equation of motion

$$(\partial_t^2 - \partial_{r^*}^2)\psi + V_\ell(r^*)\psi = 0 \quad , \tag{26.100}$$

where

$$V_\ell(r^*) = f(r)\left(\frac{\ell(\ell+1)}{r^2} + \frac{2m}{r^3}\right) \quad . \tag{26.101}$$

REMARKS:

1. Note that this potential is quite similar to the effective potential (25.43)

$$V_{eff}(r) = f(r)\frac{L^2}{2r^2} \tag{26.102}$$

for massless particles in the Schwarzschild geometry derived in section 25.3. Indeed, in the large ℓ limit the last term in (26.101) can be neglected and then V_ℓ reduces to V_{eff} with the identification $\ell(\ell+1) \to L^2$. This is as it should be and as one expects (and can show) on general grounds: in a suitable geometric optics (large ℓ, high frequency) limit the massless Klein-Gordon equation reduces to the Hamilton-Jacobi equation for null geodesics.

2. This potential is non-negative for all $2m < r < \infty$ and goes to zero quite rapidly as $r \to \infty$ or $r \to 2m$, i.e. as $r^* \to \pm\infty$,

$$\begin{aligned} r^* \to +\infty \Leftrightarrow r \to \infty: &\quad V_\ell(r) \sim (r^*)^{-2} \\ r^* \to -\infty \Leftrightarrow r \to 2m: &\quad V_\ell(r) \sim \mathrm{e}^{r^*/2m} \quad . \end{aligned} \tag{26.103}$$

This means that at infinity (in r) and near the horizon, the solutions of this equation can be chosen to have the standard right-moving (outgoing) / left-moving (ingoing) form

$$\psi(t, r^*) \sim \mathrm{e}^{\pm i\omega(t-r^*)} = \mathrm{e}^{\pm i\omega u} \quad \text{or} \quad \psi(t, r^*) \sim \mathrm{e}^{\pm i\omega(t+r^*)} = \mathrm{e}^{\pm i\omega v} \quad . \tag{26.104}$$

3. However, this does not mean that a mode having the above form near infinity, say, evolved from a mode that had also had such a form near the horizon. Rather, the almost infinite exponential gravitational redshift between the near-horizon and asymptotic regions discussed in section 26.5 leads to an exponential relation between the parameter u, say, labelling an outgoing wave at infinity, and the corresponding parameter near the horizon. This exponential relation is analogous to that encountered for a scalar field in Rindler versus inertial Minkowski coordinates (section 7.8). For the Schwarzschild metric it is encoded in the precisely analogous relation (27.156) between the coordinates u, v and the Kruskal coordinates u_K, v_K to be introduced below, the latter being the analogues of Minkowski inertial coordinates, and the former the analogues of Rindler coordinates. These observations are at the heart of the so-called *Hawking Effect*, i.e. the quantum radiation of black holes. See the references given in section 27.7 for introductions to these topics.

4. By separating out the time-dependence,

$$\psi(t, r^*) = \mathrm{e}^{-i\omega t}\psi(r^*) \ , \tag{26.105}$$

the exact equation to be solved takes the form of a standard time-independent Schrödinger equation,

$$-\partial_{r^*}^2\psi + V_\ell(r^*)\psi = \omega^2\psi \ . \tag{26.106}$$

It plays an important role in numerous aspects of Black Hole physics, e.g. in the analysis of the stability of the Schwarzschild solution. In this context, the above equation and its counterparts for vectors and symmetric tensors are known as the Regge-Wheeler(-Zerilli) equations.

27 Black Holes II: the Schwarzschild Black Hole

27.1 Preliminary Remarks

The primary purpose of this section is to understand the significance and physics of the Schwarzschild radius and the region $r < 2m$. We will acccomplish this principally via a construction of appropriate, physically motivated coordinate systems and, indeed, a secondary purpose of this section is to illustrate how to go about constructing such coordinate systems in a systematic way (instead of just introducing them without further explanations).

Even though the details will differ, the general principles of how to construct coordinates to explore and understand a given space-time (by constructing and using coordinates adapted to preferred classes of observers or geodesics) can be applied to other metrics, e.g. some of those listed in section 30.

Let me also make clear from the outset what the issue is *not*. Namely, the issue is not one of constructing appropriate coordinates solely in the region $0 < r < 2m$. Indeed, we have known such coordinates all along, simply the original Schwarzschild coordinates (t, r). The Schwarzschild metric is a vacuum solution of the Einstein equations also in that region, and the coordinates (t, r) give a valid non-singular description of the metric there. Since $f(r) = 1 - 2m/r < 0$, their interpretation differs, i.e. r is a timelike coordinate, and t plays the role of a radial coordinate, but this notational issue can easily be rectified by renaming $r = T, t = R$, and writing the Schwarzschild metric in the region $0 < r = T < 2m$ as

$$\begin{aligned} & f(r) = 1 - \frac{2m}{r} \equiv -(\frac{2m}{T} - 1) \\ \Rightarrow \quad & ds^2 = -(\frac{2m}{T} - 1)^{-1} dT^2 + (\frac{2m}{T} - 1) dR^2 + T^2 d\Omega_2^2 \ . \end{aligned} \tag{27.1}$$

While this provides some minimal insight (e.g. that for $r < 2m$ surfaces of constant r are spacelike and that the metric looks time-dependent), what we are looking for is a way of describing the physics of the Schwarzschild solution that encompasses both the region $r > 2m$ and the region $r < 2m$ (and that therefore e.g. provides a valid continuous map for the freely falling observer as he crosses $r = 2m$).

The way we will go about this is to use either the worldlines of ingoing timelike geodesic observers (along with their proper time) or those of ingoing lightrays (along with an affine parameter along them), to provide us with coordinates in the region $r < 2m$ (recall that in both cases $r = 2m$ lies at a finite value of that affine parameter so that this affine parameter is also a good coordinate beyond $r = 2m$).

We will explore the 1st option in sections 27.2 and 27.3 and the 2nd option in section 27.4 and subsequent sections. The latter, based on null geodesics, is technically somewhat simpler than the former (and is therefore also the approach commonly adopted in

the literature), and will lead us rather painlessly and quickly to the maximal analytic extension of the Schwarzschild geometry in section 27.8.

I have also included the former option here, based on timelike goedesics, precisely because the resulting coordinate systems, which are quite interesting in their own right and are useful for certain more advanced applications, are usually not dealt with in any detail in the standard textbooks (and I therefore had to work out many of these details myself at some point). However, it is possible to skip sections 27.2 and 27.3 and go directly to section 27.4 and continue from there (and this is what I usually do in a first course on General Relativity).

27.2 Crossing r_s with Painlevé-Gullstrand Coordinates

We had seen that the Schwarzschild time coordinate t is adapted to static observers (whose proper time is proportional to t), and therefore not useful for describing the region $r < 2m$. We had also seen that freely falling observers cross $r = 2m$ in finite proper time τ. This suggests to choose some family of freely falling observers (geodesics) and to use their proper time $\tau = T(t, r, \theta, \phi)$ as the new time coordinate, i.e. to perform the coordinate transformation

$$(t, r, \theta, \phi) \to (T, r, \theta, \phi) \tag{27.2}$$

(retaining, for the time being, the coordinates (r, θ, ϕ)).

A natural (and the simplest) choice is to consider the family of freely falling observers which fall radially (angular momentum $L = 0$) and start off at rest from infinity (energy $E = 1$). In this case, the geodesic equations (26.20) and (26.21) take the form

$$\dot{r} = -(2m/r)^{1/2} \quad , \quad \dot{t} = (1 - 2m/r)^{-1} \quad , \tag{27.3}$$

which also imply that

$$dt/dr = -(r/2m)^{1/2}(1 - 2m/r)^{-1} \quad . \tag{27.4}$$

The solutions for $r = r(\tau)$ and $t = t(r)$ are

$$\begin{aligned} r(\tau) &= [3\sqrt{2m}(\tau_0 - \tau)/2]^{2/3} \quad \Leftrightarrow \quad \tau - \tau_0 = -(2/3)(2m)^{-1/2} r^{3/2} \\ t(r) &= t_0 - (2/3)(2m)^{-1/2} r^{3/2} - \Theta(r) \quad , \end{aligned} \tag{27.5}$$

where

$$\Theta(r) = 2m \left(2(r/2m)^{1/2} + \ln \left| \frac{(r/2m)^{1/2} - 1}{(r/2m)^{1/2} + 1} \right| \right) \tag{27.6}$$

is the solution of

$$d\Theta(r)/dr = (2m/r)^{1/2}(1 - 2m/r)^{-1} \quad . \tag{27.7}$$

In particular, we see that, up to the irrelevant constant $\tau_0 - t_0$, $\tau(r)$ and $t(r)$ are related by

$$\tau(r) = t(r) + \Theta(r) \quad . \tag{27.8}$$

We are thus led to introduce the new time coordinate $T(t, r)$ by

$$T(t,r) = t + \Theta(r) = t + 2(2mr)^{1/2} + 2m \ln \left| \frac{(r/2m)^{1/2} - 1}{(r/2m)^{1/2} + 1} \right| \quad . \tag{27.9}$$

Since the metric does not depend explicitly on t, to determine the metric in the coordinates (T, r, θ, ϕ) we only need to substitute dt by

$$dt = dT - d\Theta = dT - (r/2m)^{-1/2}(1 - 2m/r)^{-1} dr \quad , \tag{27.10}$$

or

$$dt = dT - \frac{\sqrt{r/2m}}{(r/2m) - 1} dr \quad . \tag{27.11}$$

Then one immediately finds the simple result

$$\begin{aligned} ds^2 &= -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \\ &= -f(r)dT^2 + 2\sqrt{2m/r}dTdr + (dr^2 + r^2 d\Omega^2) \\ &= -dT^2 + \left(dr + \sqrt{2m/r}dT\right)^2 + r^2 d\Omega^2 \quad . \end{aligned} \tag{27.12}$$

This is the Schwarzschild metric in *Painlevé-Gullstrand Coordinates* (Painlevé (1921), Gullstrand (1922)), abbreviated to PG coordinates in the following, and this form of the metric already reveals a number of important properties of the Schwarzschild geometry.

Most importantly, we see that due to the non-singular off-diagonal term the metric in these coordinates is well-defined and non-degenerate for all $0 < r < \infty$, in particular at $r = 2m$. This is the definitive proof that the singularity at $r = 2m$ in Schwarzschild coordinates is really just a coordinate singularity.

REMARKS:

1. The metric has the characteristic property that the metric induced on the slices of constant T is just the flat Euclidean metric for any T,

$$ds^2|_{T=T_1} = dr^2 + r^2 d\Omega^2 = d\vec{x}^2 \quad . \tag{27.13}$$

   In particular, even though the radial coordinate distance of the coordinate r is most certainly not the proper radial distance at constant Schwarzschild time t, $ds = f(r)^{-1/2} dr \neq dr$, the above form of the metric shows that the coordinate r has the property that it does measure the proper radial distance on surfaces of constant T. In this sense, r appears to be (perhaps somewhat surprisingly) adapted to freely falling observers with $E = 1$.

In any case, this makes this form of the metric (and choice of time coordinate) particularly convenient e.g. for the canoncial quantisation of fields in the Schwarzschild space-time, and it is also for this reason that this coordinate system has become increasingly popular in recent years.[81]

2. It is easy to verify / confirm directly in the PG coordinates that the new time-coordinate T really has the interpretation as measuring the proper time τ of radially freely falling observers starting off at rest at infinity. To that end note that in PG coordinates a radial timelike geodesic satisfies

$$-\dot{T}^2 + \left(\dot{r} + \sqrt{2m/r}\dot{T}\right)^2 = -1 \quad , \tag{27.14}$$

while the conserved energy associated with the T-translation invariance of the metric has the form

$$E = f\dot{T} - \sqrt{2m/r}\dot{r} \quad . \tag{27.15}$$

It immediately follows that

$$T = \tau \quad \Rightarrow \quad \dot{T} = +1 \quad \Rightarrow \quad \dot{r} = -\sqrt{2m/r} \quad \Rightarrow \quad E = 1 \quad . \tag{27.16}$$

These are precisely the geodesic paths with which we began the construction.

The coordinate transformation (27.9) is of the general form $T(t,r) = t + \psi(r)$ (24.2) discussed previously, and preserves the t-independence and manifest spherical symmetry of the metric. It leads to the metric in the form anticipated in (24.8).

Turning this around, once one has found the Schwarzschild metric in Schwarzschild coordinates, say, one can of course "discover" an infinite number of new coordinate systems for the Schwarzschild metric by performing such a (or a more general) coordinate transformation. Only in special cases, however, will one find a coordinate system that is actually useful. Let us see how to recover the PG coordinate system in this way, and how to "precover" another coordinate system that we will discuss in more detail in section 27.4.

Starting with the Schwarzschild metric in Schwarzschild coordinates, and performing the coordinate transformation $t \to T(t,r) = t + \psi(r)$, one finds the metric

$$\begin{aligned} ds^2 &= -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \\ &= -f(r)(dT - \psi'(r)dr)^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \\ &= -f(r)dT^2 + 2f(r)\psi'(r)dTdr + f(r)^{-1}(1 - f(r)^2(\psi'(r))^2)dr^2 + r^2 d\Omega^2 \\ &\equiv -f(r)dT^2 + 2C(r)dTdr + f(r)^{-1}(1 - C(r)^2)dr^2 + r^2 d\Omega^2 \quad , \end{aligned} \tag{27.17}$$

[81] See e.g. P. Kraus, F. Wilczek, *A Simple Stationary Line Element for the Schwarzschild Geometry, and Some Applications*, `arXiv:gr-qc/9406042` and citations thereto; A. Nielsen, M. Visser, *Production and decay of evolving horizons*, `arXiv:gr-qc/0510083` for PG-like metrics for general time-dependent spherically symmetric metrics; C. Barcelo, S. Liberati, M. Visser, *Analogue Gravity*, `arXiv:gr-qc/0505065` for a detailed discussion of PG-like metrics in the context of so-called analogue models of gravity.

where

$$C(r) = f(r)\psi'(r) \tag{27.18}$$

is an essentially arbitrary function of r. This metric represents the Schwarzschild metric for any choice of $C(r)$. In particular, the vacuum Einstein equations do not impose any constraints on $C(r)$ - this can be checked explicitly, but it would be silly to do so since we have just seen explicitly that $C(r)$ is not determined and just corresponds to the freedom of performing a particular class of coordinate transformations. One can therefore now choose $C(r)$ at will.

1. One natural choice is

$$g_{Tr} = 0 \quad \Leftrightarrow \quad C(r) = 0 \ . \tag{27.19}$$

This makes the metric diagonal, corresponds to $\psi(r)$ constant, and evidently returns one to Schwarzschild coordinates. Any other choice of $C(r)$ will lead to a non-diagonal metric in the coordinates (T, r).

2. Another attractive choice is

$$\begin{aligned} g_{rr} = 1 \quad \Leftrightarrow \quad ds^2|_{T=T_0} = dr^2 + r^2 d\Omega^2 \quad &\Leftrightarrow \quad 1 - C(r)^2 = f(r) \\ &\Leftrightarrow \quad C(r) = \pm\sqrt{2m/r} \end{aligned} \tag{27.20}$$

We see that with the upper sign this is precisely the choice giving rise to Painlevé-Gullstrand coordinates introduced at the beginning of this section, confirmed by the fact that for $\psi(r)$ this implies the differential equation

$$C(r) = +\sqrt{2m/r} \quad \Rightarrow \quad \psi'(r) = \sqrt{2m/r} f(r)^{-1} \quad \Rightarrow \quad \psi(r) = \Theta(r) \ . \tag{27.21}$$

The same argument shows that the lower sign gives a corresponding set of PG coordinates based on outgoing rather than on ingoing geodesics.

3. Yet another appealing choice is

$$g_{rr} = 0 \quad \Leftrightarrow \quad C(r) = \pm 1 \quad \Leftrightarrow \quad \psi'(r) = \pm f(r)^{-1} \ , \tag{27.22}$$

so that the metric has the particularly simple (and non-singular at $r = 2m$) form

$$ds^2 = -f(r)dT^2 \pm 2dTdr + r^2 d\Omega^2 \ . \tag{27.23}$$

Referring back to the discussion of section 26.7, in particular (26.91), we see that the solution to this equation is

$$\psi(r) = \pm r^* \quad \Rightarrow \quad T(t, r) = t \pm r^* \ . \tag{27.24}$$

These are just the advanced and retarded time coordinates (u, v) of section 26.8, and we will encounter them in section 27.4 as Eddington-Finkelstein coordinates.

4. Finally, the Kerr-Schild coordinates to be discussed in section 27.6 can be obtained by the simple choice

$$C(r) = 1 - f(r) \quad \Rightarrow \quad T(t,r) = t + r^* - r = t + 2m\log(r/2m - 1) \ . \tag{27.25}$$

This procedure can in principle be applied to any static, spherically symmetric metric, and we will also make use of it later, e.g. in section 39.2 when constructing coordinates for de Sitter space.

A geometrically more satisfactory construction of such adapated coordinates can be based on the attempt to find a function T such that the 4-velocity of a family (congruence) of observers u^α is orthogonal to the slices of constant T, in the sense that $u_\alpha = -\partial_\alpha T$. The two conditions $u_\alpha u^\alpha = -1$ (for a timelike congruence, say) and $u_\alpha = -\partial_\alpha T$ imply that u^α is geodesic,

$$u^\alpha \nabla_\alpha u_\beta = u^\alpha \nabla_\beta u_\alpha = \tfrac{1}{2}\nabla_\beta(u^\alpha u_\alpha) = 0 \ , \tag{27.26}$$

(this is a special case of the general result established in (9.65) that a gradient vector field is geodesic iff it is of constant length), and that the metric component g^{TT} with respect to this new coordinate function T is

$$g^{TT} = g^{\alpha\beta}\partial_\alpha T \partial_\beta T = g^{\alpha\beta} u_\alpha u_\beta = -1 \ , \tag{27.27}$$

and the construction can proceed from there.[82]

Conversely, this kind of reasoning may allow one to discover the geometric interpretation of a coordinate system that one has selected through other criteria, e.g. via a convenient choice of the function $C(r)$ in (27.17). In particular, applied to PG coordinates, one can argue as follows:

1. Given the metric in PG coordinates, one seeks the interpretation of the gradient covector

$$u_\alpha = -\partial_\alpha T \ , \tag{27.28}$$

which is correctly normalised for a 4-velocity field, $u_\alpha u^\alpha = -1$, because $g^{TT} = -1$ in PG coordinates;

2. in Schwarzschild coordinates, u_α has the components

$$(-\partial_t T - \partial_r T, -\partial_\theta T, -\partial_\phi T) = (-1, -(r/2m)^{-1/2} f(r)^{-1}, 0, 0) \ ; \tag{27.29}$$

[82] See K. Martel, E. Poisson, *Regular coordinate systems for Schwarzschild and other spherical spacetimes*, `arXiv:gr-qc/0001069`, which also introduces generalised PG coordinates adapted to geodesic observers with $E > 1$, i.e. with non-zero velocity at infinity. Corresponding coordinates for $E < 1$, i.e. adapted to freely falling observers with maximal radius $r_i < \infty$, have been constructed by Gautreau and Hoffmann (see the references in this article).

3. thus its contravariant components give the 4-velocity field

$$u^\alpha = (f(r)^{-1}, -(r/2m)^{-1/2}, 0, 0) \ , \tag{27.30}$$

and comparison with (27.3) shows that this is precisely the family of tangent vectors

$$u^\alpha = \dot{x}^\alpha = (\dot{t}, \dot{r}, \dot{\theta}, \dot{\phi}) \tag{27.31}$$

characterising the radial ingoing geodesics with $E = 1$.

Even though we now have a coordinate system that extends in a non-singular way across $r = 2m$, so that $r = 2m$ is not a true singularity, this does not mean that nothing interesting at all happens at that locus. Indeed, the (legitimate) static observers at large radii $r \gg 2m$ are still waiting for an explanation for their observations, described in sections 26.4 and 26.5. Simply telling them that their coordinates are no good near $r = 2m$ will hardly be considered by them to be a satisfactory explanation of what they observe.

In order to address this issue, we now look at the behaviour of (radial) lightrays in PG coordinates, characterised by

$$-f(r)dT^2 + 2\sqrt{2m/r}dTdr + dr^2 = 0 \ . \tag{27.32}$$

Parametrising the lightrays by $r = r(T)$, with a prime denoting a T-derivative, $r' = dr/dT$, we can write this equation as

$$(r')^2 + 2\sqrt{2m/r}r' = f \quad \Leftrightarrow \quad (r' + \sqrt{2m/r})^2 = 1 \ . \tag{27.33}$$

Thus this has the two solutions

$$r' + \sqrt{2m/r} = \pm 1 \quad \Leftrightarrow \quad r'_\pm = \pm 1 - \sqrt{2m/r} \ , \tag{27.34}$$

and we can now understand the role that $r = 2m$ plays.

For $r > 2m$ one has

$$r > 2m : \quad r'_+ > 0 \quad \text{and} \quad r'_- < 0 \ . \tag{27.35}$$

This is the usual situation. There is one outgoing direction along which $r = r(T)$ grows, and one ingoing direction along which $r(T)$ decreases. In particular, while remaining within the future lightcone one can choose to go to either larger or smaller values of r.

For $r < 2m$, on the other hand, one has

$$r < 2m : \quad r'_+ < 0 \quad \text{and} \quad r'_- < 0 \ . \tag{27.36}$$

Thus along both directions lightrays (and therefore massive particles as well) must move to smaller values of r. In particular, *no observer, no lightray and no information can escape from the region* $r < 2m$. No wonder that the asymptotic static observers never

"see" the infalling observer cross $r = 2m$. This new region is a future extension of the Schwarzschild "patch" $r > 2m$ of the space-time, uncovered by ingoing radial geodesics.

With the opposite choice of sign in (27.20), corresponding to outgoing rather than ingoing radial geodesics, one would instead have discovered a region $r < 2m$ in which $r'_{\pm} > 0$. This can therefore not possibly be the "same" region $r < 2m$, and indeed is a past extension of the Schwarzschild patch, uncovered by the back-tracking of outgoing radial geodesics.

While it is possible to further explore the consequences of all this in terms of the present PG coordinates, since we have been led to consider the structure of the lightcones, i.e. null geodesics, it turns out to be more convenient, also for the following, to discuss this in terms of coordinates that are adapted to lightrays rather than to the timelike geodesics. These are the Eddington-Finkelstein coordinates to be discussed in section 27.4 below.

27.3 Lemaître and Novikov Coordinates

Before turning to Eddington-Finkelstein coordinates, I want to briefly mention two more coordinate systems that are instructive and occasionally useful, and that combine one of the attractive feature of PG coordinates (namely that the time coordinate is physical proper time for suitable geodesic observers) with one of the useful features of Schwarzschild coordinates (namely that the metric is diagonal), albeit at the expense of a time-dependent coordinate transformation rendering also the metric time-dependent.

This will be accomplished by constructing *comoving coordinates* for the geodesic observers underlying the (generalised) PG coordinate system, where comoving means that these observers remain at fixed values of all the spatial coordinates and only evolve in time = proper time. Since radial geodesics already remain at fixed values of the angular coordinates (θ, ϕ), what this amounts to is to trade the coordinate r for another coordinate that simply labels the individual geodesics (and that is therefore, tautologically, guaranteed to also remain constant along such a geodesic). Such a label is provided by an integration constant appearing in the solution to the radial geodesic equation since (tauto-)logically such an integration constant is constant along the geodesic.

Let us first consider the case $E = 1$. In this case, the effective potential for radial motion gives us the radial equation

$$\dot{r}^2 = \frac{2m}{r} \quad , \tag{27.37}$$

with the ingoing solution (27.5),

$$\dot{r} = -\sqrt{2m/r} \quad \Rightarrow \quad r(\tau) = [3\sqrt{2m}(\tau_0 - \tau)/2]^{2/3} \quad . \tag{27.38}$$

We can thus immediately identify τ_0 as a candidate for a comoving radial coordinate. This label τ_0 can be interpreted as the proper time at which the geodesic ends up at

$r = 0$, $r(\tau_0) = 0$. Alternatively and preferably, if one is (understandably) reluctant to label geodesics by their behaviour at $r = 0$, τ_0 can be thought of as being related to the value r_0 of $r(\tau)$ at $\tau = 0$,

$$r_0 = r(\tau = 0) = [3\sqrt{2m}\tau_0/2]^{2/3} \quad . \tag{27.39}$$

We therefore introduce a new radial variable $\rho = \rho(r, \tau)$ by

$$r(\tau, \rho) = [3\sqrt{2m}(\rho - \tau)/2]^{2/3} \quad . \tag{27.40}$$

This satisfies

$$dr = \dot{r}d\tau + (\partial r/\partial\rho)d\rho = \dot{r}(d\tau - d\rho) = -\sqrt{2m/r}(d\tau - d\rho) \quad , \tag{27.41}$$

so that (denoting the PG proper time coordinate T now by τ) the Schwarzschild line element in PG form (27.12) turns into

$$\begin{aligned} ds^2 &= -d\tau^2 + \left(dr + \sqrt{2m/r}d\tau\right)^2 + r^2 d\Omega^2 \\ &= -d\tau^2 + \left(\sqrt{2m/r}d\rho\right)^2 + r^2 d\Omega^2 \\ &= -d\tau^2 + \frac{2m}{r(\tau,\rho)}d\rho^2 + r(\tau,\rho)^2 d\Omega^2 \quad , \end{aligned} \tag{27.42}$$

with $r(\tau, \rho)$ given explicitly by (27.40). This is the Schwarzschild metric in *Lemâitre Coordinates* (Lemaître (1938)).

REMARKS:

1. These coordinates have a clear physical interpretation and also manifestly extend in a non-singular way across $r = 2m$, the Schwarzschild radius being located at the innocuous value
$$r = 2m \quad \Rightarrow \quad \rho - \tau = 4m/3 \tag{27.43}$$
of the Lemaître coordinates, to all $r > 0$.

2. This is the first (but will not be the last) time that we see that it can be useful to work with a radial coordinate, here $\rho = \rho(r, \tau)$, that is not equal to the standard (aereal radius) radial coordinate r.

3. For comparison purposes with the metric in Novikov coordinates to be discussed below, note that from the explicit change of variables (27.40) one has
$$\frac{2m}{r(\tau,\rho)} = (\partial_\tau r)^2 = (\partial_\rho r)^2 \quad , \tag{27.44}$$
so that the metric (27.42) can also be written as
$$\begin{aligned} ds^2 &= -d\tau^2 + \frac{2m}{r(\tau,\rho)}d\rho^2 + r(\tau,\rho)^2 d\Omega^2 \\ &= -d\tau^2 + (\partial_\rho r)^2 d\rho^2 + r(\tau,\rho)^2 d\Omega^2 \quad . \end{aligned} \tag{27.45}$$

This also shows that the form of the metric does not depend on the precise choice of integration constant used to label the geodesics, since it is manifestly invariant under transformations

$$\rho \to \sigma = F(\rho) \quad \Rightarrow \quad (\partial_\rho r)^2 d\rho^2 = (\partial_\sigma r)^2 d\sigma^2 \ . \tag{27.46}$$

4. Since

$$dr = 0 \quad \Rightarrow \quad d\tau = d\rho \ , \tag{27.47}$$

the metric induced on surfaces of constant $r = r_1$ is given by

$$ds^2|_{r=r_1} = -\left(1 - \frac{2m}{r_1(\tau,\rho)}\right) d\tau^2 + r_1(\tau,\rho)^2 d\Omega^2 \ . \tag{27.48}$$

Thus, as could have (partially) been anticipated from the Schwarzschild form of the metric, a hypersurface of constant $r = r_1$ is timelike for $r_1 > 2m$, null for $r_1 = 2m$, and spacelike for $r_1 < 2m$.

5. The surfaces of constant time $\tau = \tau_1$, on the other hand, are manifestly spacelike everywhere,

$$ds^2|_{\tau=\tau_1} = \frac{2m}{r(\tau_1,\rho)} d\rho^2 + r(\tau_1,\rho)^2 d\Omega^2 \ . \tag{27.49}$$

They run into the spacelike singularity at $r = 0$ at $\rho = \tau_1$.

6. Since from (27.41) one sees that at constant $\tau = \tau_1$ one has

$$\sqrt{\frac{2m}{r}} d\rho|_{\tau=\tau_1} = dr \ , \tag{27.50}$$

this spatial metric induced on surfaces of constant τ is again, as in the case of PG coordinates (27.13), just the Euclidean metric on $\mathbb{R}^3$,

$$ds^2|_{\tau=\tau_1} = dr^2 + r^2 d\Omega^2 \ . \tag{27.51}$$

This should not come as a surprise since the PG coordinate T is equal to the Lemaître coordinate τ.

7. In these coordinates, the volume element $\sqrt{g}$ has the simple form

$$\sqrt{g} = \sqrt{2m} r^{3/2} \sin\theta = 3m(\rho - \tau)\sin\theta \ . \tag{27.52}$$

8. Even though the metric is explicitly time-dependent in Lemaître coordinates, the time-translation invariance of the Schwarzschild metric is still manifest in these coordinates. Indeed, since $r(\tau,\rho)$ only depends on the difference $\rho - \tau$, the metric is invariant under simultanous constant translations

$$(\tau,\rho) \to (\tau + c, \rho + c) \tag{27.53}$$

of τ and ρ. Equivalently, in Lemaître coordinates the Killing vector ∂_t takes the form

$$\partial_t = \partial_\tau + \partial_\rho \ . \tag{27.54}$$

The norm of this Killing vector is

$$g_{\alpha\beta}(\partial_t)^\alpha(\partial_t)^\beta = -(1 - \frac{2m}{r}) \ , \tag{27.55}$$

and thus ∂_t is timelike for $r(\tau,\rho) > 2m$, null for $r = 2m$, and spacelike for $0 < r < 2m$.

9. Note that $g_{\tau\tau} = -1$ and $g_{\tau\rho} = 0$ are automatic consequences of this construction: by definition, in comoving coordinates the velocity field has the form $u^\alpha = (1,0,0,0)$, and $u_\alpha = -\partial_\alpha\tau$ implies

$$\begin{aligned} u^\alpha = -g^{\alpha\beta}\partial_\beta\tau = -g^{\alpha\tau} \stackrel{!}{=} (1,0,0,0) \quad &\Rightarrow \quad g^{\tau\tau} = -1 \quad , \quad g^{\tau\rho} = 0 \\ &\Rightarrow \quad g_{\tau\tau} = -1 \quad , \quad g_{\tau\rho} = 0 \ . \end{aligned} \tag{27.56}$$

We will discuss comoving coordinates, and metrics employing them, in much more detail in the context of cosmology in sections 33-38, cf. in particular section 34.2 for geodesics of comoving observers in comoving coordinates,

10. Somewhat surprisingly, Lemaître coordinates appear not to have attracted a lot of attention or found widespread use. However, the fact that in Lemaître coordinates the metric extends all the way to $r = 0$, is diagonal, and is explicit (unlike the Novikov coordinates, briefly discussed below, or the Kruskal-Szekeres coordinates to be described in detail later on) again makes them an attractive coordinate system to use e.g. when studying quantum field theory in a Schwarzschild background.[83]

Closely related to Lemaître coordinates are the so-called *Novikov coordinates*, comoving coordinates based on radial geodesics with a finite maximal radius, $r_i < \infty$ (i.e. $E < 1$) in the notation of section 26.3. In this case, the equation to solve is

$$\dot{r}^2 = \frac{2m}{r} + 2E_{eff} = \frac{2m}{r} + E^2 - 1 \ , \tag{27.57}$$

where E_{eff} or E depends on the choice of geodesic via a choice of integration constant (one could even choose E itself, say, as a comoving coordinate). Using r_i instead, related to the energy by

$$E(r_i)^2 = 1 - \frac{2m}{r_i} = f(r_i) = 1 + 2E_{eff}(r_i) \ , \tag{27.58}$$

[83] See e.g. K. Melnikov, M. Weinstein, *On Unitary Evolution of a Massless Scalar Field In A Schwarzschild Background: Hawking Radiation and the Information Paradox*, `arXiv:hep-th/0205223`.

the trajectories are implicitly defined by (26.28) and (26.29), i.e.

$$\begin{aligned} r(\eta) &= \tfrac{1}{2} r_i \, (1+\cos\eta) \\ \tau(\eta) &= \left(\frac{r_i^3}{8m}\right)^{1/2} (\eta + \sin\eta) \quad . \end{aligned} \tag{27.59}$$

We now think of these relations, together with the usual relation $E = f\dot{t}$ (26.21) as defining a change of variables

$$r = r(\tau, r_i) \quad , \quad t = t(\tau, r_i) \quad . \tag{27.60}$$

Note that r_i is clearly a comoving coordinate, as it can be used to label the geodesic. Note also that, if required/desired, from (27.59) one can solve for $\tau = \tau(r, r_i)$,

$$\begin{aligned} & \eta = \arccos(2r/r_i - 1) \\ \Rightarrow \quad & \tau(r, r_i) = \left(\frac{r_i^3}{8m}\right)^{1/2} \left(\arccos(2r/r_i - 1) + 2\sqrt{(r/r_i) - (r/r_i)^2}\right) \quad . \end{aligned} \tag{27.61}$$

However, even without having to solve or invert these implicit equation, it is easy to see that the resulting metric will have the form

$$ds^2 = -d\tau^2 + f(r_i)^{-1} r'(\tau, r_i)^2 dr_i^2 + r(\tau, r_i)^2 d\Omega^2 \quad , \tag{27.62}$$

where $r'(\tau, r_i) = \partial r(\tau, r_i)/\partial r_i$. This is the metric in *Novikov Coordinates* (Novikov, 1964).

In order to establish (27.62), we proceed as follows:

- Writing
$$dt = \dot{t} d\tau + t' dr_i \quad , \quad dr = \dot{r} d\tau + r' dr_i \quad , \tag{27.63}$$
and plugging this into the Schwarzschild metric, one finds
$$ds^2 = -(f\dot{t}^2 - f^{-1}\dot{r}^2) d\tau^2 + 2(f^{-1}\dot{r}r' - f\dot{t}t') d\tau dr_i + (f^{-1}(r')^2 - f(t')^2) dr_i^2 \quad . \tag{27.64}$$

- The first term in brackets is equal to 1 because $(t(\tau), r(\tau))$ parametrise a timelike radial geodesic,
$$\text{timelike radial geodesic} \quad \Rightarrow \quad -f(r)\dot{t}^2 + f(r)^{-1}\dot{r}^2 = -1 \quad . \tag{27.65}$$
Thus $g_{\tau\tau} = -1$, as expected for comoving coordinates.

- Moreover, for comoving coordinates the second term in brackets, i.e. the off-diagonal term, is zero, $g_{\tau r_i} = 0$ (27.56). Thus we can deduce
$$g_{\tau r_i} = 0 \quad \Rightarrow \quad f^{-1}\dot{r}r' - f\dot{t}t' = 0 \quad . \tag{27.66}$$

Using

$$\dot{r} = -(f_i - f)^{1/2} \quad , \quad f\dot{t} = E = f_i^{1/2} \qquad (f_i \equiv f(r_i)) \ , \tag{27.67}$$

this allows us to eliminate t' in favour of r',

$$(t')^2 = f^{-2}(1 - f/f_i)(r')^2 \ . \tag{27.68}$$

- This implies for the third term in brackets that

$$g_{r_i r_i} = f^{-1}(r')^2 - f(t')^2 = f_i^{-1}(r')^2 \ , \tag{27.69}$$

leading to the result given in (27.62).

Remarks:

1. Note that, just as the Lemaître metric in (27.45), the form of the Novikov metric does not depend on the precise choice of integration constants used to label the geodesics, as the term $(\partial_{r_i} r)^2 (dr_i)^2$ is invariant under transformations

$$r_i \to \rho = F(r_i) \ . \tag{27.70}$$

Both metrics can then uniformly be presented as

$$ds^2 = -d\tau^2 + \frac{(\partial_\rho r)^2}{1 + 2E_{eff}(\rho)} d\rho^2 + r(\tau, \rho)^2 d\Omega^2 \ , \tag{27.71}$$

where $r(\tau, \rho)$ is the solution to the radial equation of motion

$$\left(\frac{\partial r}{\partial \tau}\right)^2 = \frac{2m}{r} + 2E_{eff}(\rho) \ . \tag{27.72}$$

For $E_{eff}(\rho) = 0$ this reduces to the metric in Lemaître coordinates, for $E_{eff} < 0$ and $\rho = r_i$ this reduces to the above metric in Novikov coordinates, and one can likewise consider Novikov coordinates based on radial geodesics with $E_{eff} > 0$, i.e. with a non-zero velocity at infinity.

2. Usually, the metric is expressed not in terms of the variable r_i but in terms of

$$R = (r_i/2m - 1)^{1/2} \quad \Leftrightarrow \quad r_i = 2m(R^2 + 1) \quad \Leftrightarrow \quad E^2 = f_i = \frac{R^2}{R^2 + 1} \tag{27.73}$$

so that the metric takes the form

$$ds^2 = -d\tau^2 + \frac{R^2 + 1}{R^2} \left(\frac{\partial r}{\partial R}\right)^2 dR^2 + r(\tau, R)^2 d\Omega^2 \ . \tag{27.74}$$

In terms of R, the relation (27.61) can also, using

$$\arccos(2r/r_i - 1) = 2 \arccos(r/2m(R^2 + 1))^{1/2} \ , \tag{27.75}$$

which in turn can be established by using the trigonometric identity

$$\cos 2\varphi = 2\cos^2\varphi - 1 \quad , \tag{27.76}$$

be written as

$$\frac{\tau(r,R)}{2m} = (R^2+1)\left[\frac{r}{2m} - \frac{(r/2m)^2}{R^2+1}\right]^{1/2} + (R^2+1)^{3/2}\arccos\left[\left(\frac{r/2m}{R^2+1}\right)^{1/2}\right] \quad . \tag{27.77}$$

This expression is occasionally found and used in the literature.

3. The fact that $r(\tau, r_i)$ or $r(\tau, R)$ is only determined implicitly makes Novikov coordinates somewhat more awakward to use in practice than Lemaître coordinates.[84] Nevertheless, this is compensated by their clear physical interpretation. In particular, they are useful e.g. in numerical simulations and other investigations of gravitational collapse, and the Novikov metric will indeed naturally arise in this context in our brief discussion of gravitational collapse in section 29, in particular first in section 29.3.

4. Since the timelike geodesics that Novikov coordinates are based on oscillate in finite proper time from $r = 0$ in the past through the maximal radius r_i to $r = 0$ in the future, Novikov coordinates cover both the past and future extensions of the metric in the Schwarzschild patch discovered in terms of PG coordinates in section 27.2, and to be discussed again below in section 27.5. The reflection symmetry $R \to -R$ of the metric also exchanges the Schwarzschild patch and its "mirror region" (first encountered in the context of isotropic coordinates at the end of section 24.5, and to be discussed in detail in section 27.8), and thus Novikov coordinates actually turn out to provide a complete covering of the fully extended Kruskal-Schwarzschild space-time.[85]

27.4 Eddington-Finkelstein Coordinates I: Regularity of the Metric

In previous sections we have used the worldlines of freely falling observers and in particular the proper time along such worldlines, to construct (in various ways) coordinates in which the Schwarzschild metric extends in a non-singular way across $r = 2m$.

Instead of singling out some class of timelike observers in order to construct coordinates, it is at least equally (if not more) natural to introduce coordinates that are adapted to null geodesics.

[84] See e.g. the appendix of J. Makela, A. Peltola, *Thermodynamical Properties of Horizons*, `arXiv:gr-qc/0205128` for somewhat more explicit expressions, including a Taylor expansion of r'.

[85] See e.g. Figure 31.2 of C. Misner, K. Thorne, J. Wheeler, *Gravitation*.

We can easily accomplish this by promoting the integration constants $C_\pm$ in (26.93) labelling the lightray to new coordinates, namely the retarded or advanced time coordinate

$$C_+ \to u = t - r^* \quad , \quad C_- \to v = t + r^* \ , \tag{27.78}$$

where

$$r^* = r + 2m \log |r/2m - 1| \tag{27.79}$$

is the solution of

$$\frac{dr^*}{dr} = f(r)^{-1} \ . \tag{27.80}$$

Then ingoing radial null geodesics ($dr^*/dt = -1$) are characterised by $v = const.$ and outgoing radial null geodesics by $u = const.$ (and u and v can be thought of as "comoving" coordinates for outgoing resp. ingoing lightrays).

Then we can label space-time points (in addition to by their angular coordinates) e.g. in terms of ingoing lightrays by specifying the lightray (i.e. v) and the affine parameter λ indicating how far one has to travel along that lightray to reach the point. As we had seen explicitly e.g. in (26.82), this affine parameter can conveniently be chosen to be the radial coordinate r itself, and evidently extends across $r = 2m$.

Thus we now pass to the ingoing or outgoing *Eddington-Finkelstein* coordinates (v, r, θ, ϕ) or (u, r, θ, ϕ) (i.e. we keep r but eliminate t). Note that this coordinate transformation $u(t, r) = t - r^*$ or $v(t, r) = t + r^*$ is also of the general form $T(t, r) = t + \psi(r)$ whose consequences were already explored in section 27.2. Since the metric does not depend explicitly on t, we only need to substitute dt using

$$dt = dv - dr^* = dv - dr/f(r) \tag{27.81}$$

(and likewise for u). It follows that in terms of these coordinates the Schwarzschild metric reads

$$ds^2 = -(1 - 2m/r)dv^2 + 2dv\ dr + r^2 d\Omega^2 \tag{27.82}$$

in ingoing (advanced) Eddington-Finkelstein coordinates, and

$$ds^2 = -(1 - 2m/r)du^2 - 2du\ dr + r^2 d\Omega^2 \tag{27.83}$$

in outgoing (retarded) Eddington-Finkelstein coordinates, as already anticipated in (27.23). This way of writing the Schwarzschild metric turns out to be extremely informative and useful. We will focus on the metric and its regularity in this section, and discuss the causal structure (lightcones) and its implications in section 27.5 below.

REMARKS:

1. Even though the metric coefficent g_{uu} or g_{vv} vanishes at $r = 2m$, as for the PG coordinates of section 27.2 there is no real degeneracy, the two-dimensional

metric in the (v,r)- or (u,r)-directions having the completely non-singular and non-degenerate form

$$(g_{(v,r),(v,r)}) = \begin{pmatrix} -f(r) & \pm 1 \\ \pm 1 & 0 \end{pmatrix} \tag{27.84}$$

Indeed, the determinant and inverse of this (2×2)-block of the metric are

$$|\det(g_{(v,r),(v,r)})| = 1 \tag{27.85}$$

and

$$(g^{(v,r),(v,r)}) = \begin{pmatrix} 0 & \pm 1 \\ \pm 1 & f(r) \end{pmatrix} \ , \tag{27.86}$$

both of which are completely regular for all $r > 0$, in particular at $r = 2m$.

Therefore we can now extend the range of r to the region $0 < r < 2m$ with impunity. Thus this provides another explicit proof that the singularity of the Schwarzschild metric in the Schwarzschild coordinates at $r = 2m$ is a removable (pure coordinate) singularity.

2. Given the Eddington-Finkelstein coordinates, in which the metric is regular at and across $r = 2m$, it is now instructive to revisit some of the preliminary calculations that we performed in section 26 while exploring the Schwarzschild metric in the region $r \gtrsim 2m$.

   In particular, let us reconsider the calculation of the acceleration of a static observer (section 26.2). In Schwarzschild (SS) coordinates (which for the purposes of this remark we will denote by y^μ, reserving x^α for the Eddington-Finkelstein (EF) coordinates), the acceleration vector of a static oberver with 4-velocity (26.6)

$$(u^\mu)_{SS} = (u^t, u^r, u^\theta, u^\phi) = (f(r)^{-1/2}, 0, 0, 0) \tag{27.87}$$

   was found to be (26.10)

$$(a^\mu)_{SS} = (a^t, a^r, a^\theta, a^\phi) = (0, m/r^2, 0, 0) \ , \tag{27.88}$$

   which looks nicely regular and unspectacularly "Newtonian". This was misleading, however, as the norm of the acceleration vector involves the component $g_{rr} = f(r)^{-1}$ of the metric, leading to

$$g_{\mu\nu}(a^\mu)_{SS}(a^\nu)_{SS} = f(r)^{-1}(m/r^2)^2 \ , \tag{27.89}$$

   which diverges as $r \to 2m$.

   Let us now look at how this calculation presents itself in EF coordinates in which, as we have seen, the metric is regular at $r = 2m$. In EF coordinates, a static observer has 4-velocity

$$(u^\alpha)_{EF} = (u^v, u^r, u^\theta, u^\phi) = (u^v, 0, 0, 0) \ , \tag{27.90}$$

with

$$g_{\alpha\beta}(u^\alpha)_{EF}(u^\beta)_{EF} = -f(r)(u^v)^2 = -1 \quad \Rightarrow \quad u^v = f(r)^{-1/2} \ . \tag{27.91}$$

Thus

$$(u^\alpha)_{EF} = (f(r)^{-1/2}, 0, 0, 0) \ . \tag{27.92}$$

This agrees with the result in SS coordinates, as could have also been deduced from the vectorial transformation behaviour

$$(u^v)_{EF} = \frac{\partial v}{\partial y^\mu}(u^\mu)_{SS} = \frac{\partial v}{\partial t}(u^t)_{SS} = (u^t)_{SS} \ . \tag{27.93}$$

under the coordinate transformation

$$dv = dt + dr_* = dt + dr/f(r) \quad \Rightarrow \quad \frac{\partial v}{\partial t} = 1 \quad , \quad \frac{\partial v}{\partial r} = f(r)^{-1} \ . \tag{27.94}$$

The acceleration vector can now be determined

- either by calculating

$$a^\alpha = \Gamma^\alpha_{\beta\gamma} u^\beta u^\gamma = \Gamma^\alpha_{vv}(u^v)^2 = f(r)^{-1}\Gamma^\alpha_{vv} \ . \tag{27.95}$$

  in EF coordinates (Exercise!),

- or by transforming $(a^\mu)_{SS}$ to EF coordinates,

$$(a^\alpha)_{EF} = \frac{\partial x^\alpha}{\partial y^\mu}(a^\mu)_{SS} = \frac{\partial x^\alpha}{\partial r}(a^r)_{SS} \quad \Rightarrow \quad \begin{cases} (a^v)_{EF} = f(r)^{-1} m/r^2 \\ (a^r)_{EF} = m/r^2 \end{cases} \tag{27.96}$$

Either way, the result is

$$(a^\alpha)_{EF} = (a^v = f(r)^{-1} m/r^2, a^r = m/r^2, 0, 0) \ . \tag{27.97}$$

Thus the (non-singular, "Newtonian") r-component agrees with that of the acceleration in SS coordinates, but in addition in EF coordinates there is now a v-component which is singular as $r \to 2m$.

Thus the norm of the acceleration in EF coordinates is

$$g_{\alpha\beta}(a^\alpha)_{EF}(a^\beta)_{EF} = -f(r)(a^v)^2 + 2a^v a^r = f(r)^{-1}(m/r^2)^2 \ , \tag{27.98}$$

in complete agreement with the result in SS coordinates (as it should, since this is a scalar).

We see that, in these coordinates in which the metric is regular, the divergence of the acceleration shows up in the components of that quantity itself (whereas in the "singular" SS coordinates that divergence was misleadingly hidden in the metric).

3. One can of course check directly that the Eddington-Finkelstein metric (27.82) is a solution of the vacuum Einstein equations. One can also obtain the metric directly from integrating the Einstein vacuum equations if one starts not with the standard form of a static isotropic metric (24.6) but makes an ansatz of the form

$$ds^2 = -f(v,x)dv^2 + 2dv\,dx + r(v,x)^2 d\Omega^2 \tag{27.99}$$

in terms of two unknown functions $f(v,x)$ and $r(v,x)$ of two variables. The same arguments as in the discussion of ingoing null geodesics in section 27.5 below show that this general form of the metric is characterised by the fact that that the integral curves of the null vector ∂_x (i.e. the curves with constant (v,θ,ϕ)) are null geodesics, with affine parameter x,

$$g_{xx} = 0 \quad , \quad \nabla_{\partial_x}\partial_x = 0 \ . \tag{27.100}$$

As one can always choose to build a coordinate system in such a way, this is a valid a priori ansatz for the metric, and from this perspective one never encounters the question if the singularity at $r = 2m$ is real or not, since it is not even a coordinate singularity in these coordinates.

Alternatively, one can start with the metric in the Bondi gauge (40.30),

$$ds^2 = -\mathrm{e}^{2h(v,r)} f(v,r)dv^2 + 2\mathrm{e}^{h(v,r)} dv\,dr + r^2 d\Omega^2 \ . \tag{27.101}$$

This is tantamount to introducing the aereal radius $r(v,x)$ as a new coordinate instead of x, and this ansatz is the Eddington-Finkelstein-like counterpart of the Schwarzschild-Birkhoff ansatz (24.68). It is shown in section 40.3 that solving the vacuum Einstein equations in this gauge, one also obtains directly the Schwarzschild solution in Eddington-Finkelstein coordinates.

4. In the Eddington-Finkelstein coordinates, the Killing vector $\xi = \partial_t$ generating the time translation invariance of the metric in Schwarzschild coordinates can be written as

$$\xi = \partial_t = (\partial_t r)\partial_r + (\partial_t v)\partial_v = \partial_v \tag{27.102}$$

or

$$\xi = \partial_t = (\partial_t r)\partial_r + (\partial_t u)\partial_u = \partial_u \ . \tag{27.103}$$

That this is indeed a Killing vector is obvious from the fact that in Eddington-Finkelstein coordinates the components of the metric do not depend on v or u.

In particular, ξ now extends smoothly across $r = 2m$, with norm

$$g_{\alpha\beta}\xi^\alpha\xi^\beta = -(1 - 2m/r) \ . \tag{27.104}$$

Thus ξ is timelike for $r > 2m$, null on $r = 2m$ and spacelike for $0 < r < 2m$. This is a crucial and characteristic feature we will come back to on various occasions in subsequent sections.

27.5 Eddington-Finkelstein Coordinates II: Event Horizons and Black Holes

We now take a closer look at the causal structure of the extended Schwarzschild geometry in Eddington-Finkelstein coordinates. To determine the lightcones in ingoing Eddington-Finkelstein coordinates we again look at radial null geodesics which this time are solutions to

$$(1-2m/r)dv^2 = 2dv\,dr \quad . \tag{27.105}$$

There are thus two possibilities:

1. Ingoing null geodesics are described by $dv/dr = 0$ or $v = const.$. These null geodesics can also be understood in a different way. Note that from the Eddington-Finkelstein form (27.82) of the metric it is evident that ∂_r is a null vector, since $g_{rr} = 0$. Moreover, its integral curves, i.e. the curves

$$x^\alpha(\lambda) = (r = r_0 - \lambda, v = v_0, \theta = \theta_0, \phi = \phi_0) \tag{27.106}$$

   are geodesics. The sign $-\lambda$ has been chosen so that the tangent vector $\dot{x}^\alpha = dx^\alpha/d\lambda$ is future-oriented, i.e. such that its scalar product with $\partial_t = \partial_v$ is negative,

$$g_{\alpha\beta}\dot{x}^\alpha(\partial_t)^\beta = g_{\alpha v}\dot{x}^\alpha = \dot{r} \overset{!}{<} 0 \quad . \tag{27.107}$$

   Thus the radius indeed decreases along future-oriented ingoing null geodesics (as the name was meant to suggest) and the radial coordinate is an affine parameter along these geodesics (as we also already saw in (26.82)).

   That these curves are geodesics can be seen explicitly by calculating the acceleration of $u^\alpha = (0, -1, 0, 0)$ (in the coordinates (v, r, θ, ϕ)),

$$u^\alpha \nabla_\alpha u^\beta = -\nabla_r u^\beta = -\Gamma^\beta_{r\gamma} u^\gamma = \Gamma^\beta_{rr} \quad , \tag{27.108}$$

   or more concisely (see (5.21)) by

$$\nabla_{\partial_r}\partial_r = \Gamma^\beta_{rr}\partial_\beta \quad . \tag{27.109}$$

   Either way this shows that ∂_r is geodesic, as $\Gamma^\beta_{rr} = 0$ (since $g_{rr} = 0$, the only possible contribution to $\Gamma_{\beta rr}$ could have arisen from $\partial_r g_{vr}$, but $g_{vr} = 1$ is constant).

2. Outgoing null geodesics are described by the solutions to the equation

$$\frac{dv}{dr} = 2(1-2m/r)^{-1} \quad . \tag{27.110}$$

   The solution to this equation is evidently

$$v(r) = 2r^* + C \quad \Leftrightarrow \quad u = t - r^* = C \quad . \tag{27.111}$$

   We thus reassuringly recover the fact that outgoing lightrays are described by lines of constant u.

Thus the metric and the lightcones remain well-behaved (do not fold up) at $r = 2m$, the surface $r = 2m$ is at a finite coordinate distance, namely (to reiterate the obvious) at $r = 2m$, and there is no problem with following geodesics beyond $r = 2m$.

In particular, this means that we now encounter no difficulties when entering the region $r < 2m$, e.g. along lines of constant v and this region should be included as part of the physical space-time. Note that because $v = t + r^*$ and $r^* \to -\infty$ for $r \to 2m$, we see that decreasing r along lines of constant v amounts to $t \to \infty$. Thus the new region at $r \leq 2m$ we have discovered is in some sense a future extension of the original Schwarzschild space-time.

To understand the nature of this new region, we now take a more detailed look at the behaviour of the lightcones. Even though the lightcones do not fold up at $r = 2m$, something interesting is certainly happening there. Whereas, in a (v, r)-diagram (see Figure 18), one side of the lightcone always remains horizontal (at $v = const.$), the other side becomes vertical at $r = 2m$ ($dv/dr = \infty$) and then tilts over to the other side. In particular, beyond $r = 2m$ all future-directed paths, those within the forward lightcone, now have to move in the direction of decreasing r: clearly the ingoing null geodesics move towards smaller values of r, but so do those that for $r > 2m$ were outgoing,

$$(27.110) \quad \Rightarrow \quad dr/dv < 0 \quad \text{for} \quad r < 2m \ . \tag{27.112}$$

There is thus no way to turn back to larger values of r, not on a geodesic but also not on any other path (i.e. not even with a powerful rocket) once one has gone past $r = 2m$.

Thus, even though locally the physics at $r = 2m$ is well behaved, globally the surface $r = 2m$ is very significant as it is a point of no return:

- Since $r = 2m$ is a null surface, once one has reached the event horizon one has to travel at the speed of light to stay there and not be forced further towards smaller values of r.
- Once one has passed the Schwarzschild radius, there is no turning back to larger values of r.

Therefore nothing, absolutely nothing, no information, no lightray, no particle, can escape from the region $r < 2m$. Thus we have a *Black Hole*, an object that is (classically) completely invisible from the outside. In particular, no information about any events occurring in the black hole region $r < 2m$ can reach the asymptotic ($r >> 2m$) region. The boundary of this region is the surface $r = 2m$, and a (necessarily null) surface with this property is known as an *Event Horizon*.

Even though the Eddington-Finkelstein coordinates (more precisely the closely related Eddington time coordinate - cf. section 27.6 below) were already introduced by Eddington back in 1924, this full significance of the Schwarzschild radius and its interpretation as a "one-way membrane" were only understood much later (Finkelstein, 1958).

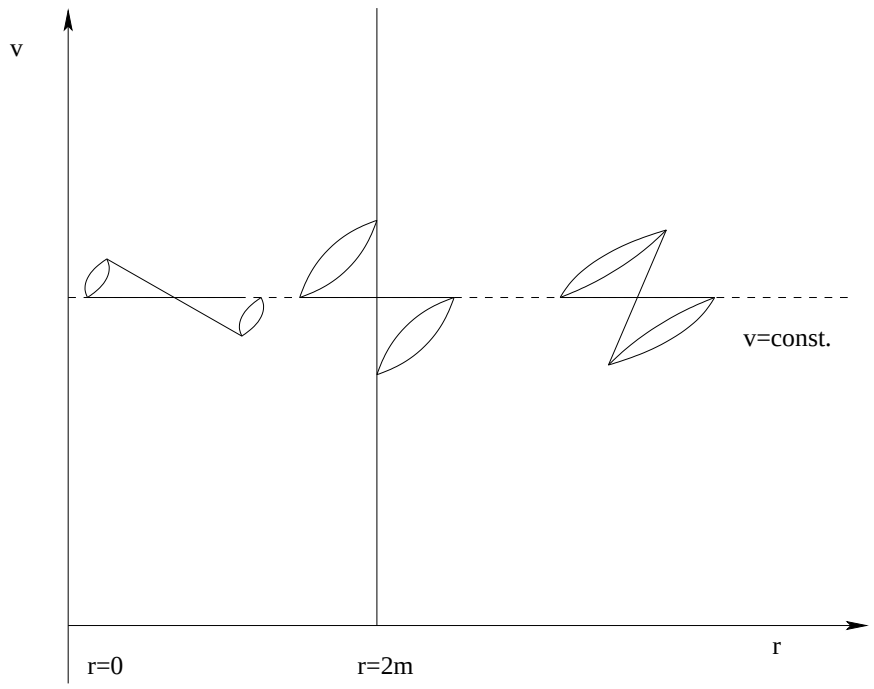


Figure 18: Behaviour of lightcones in ingoing Eddington-Finkelstein coordinates. Lightcones do not fold up at $r = 2m$ but tilt over so that for $r < 2m$ only movement in the direction of decreasing r towards the singularity at $r = 0$ is allowed.

REMARKS:

1. In the above (v, r) coordinate system we can cross the event horizon only on future directed paths, not on past directed ones, and only in the direction of decreasing r. However, clearly this cannot be the whole story: the Schwarzschild metric in the Schwarzschild coordinates is invariant under time-reflections $t \to -t$. Hence there must also exist a time-reversed version of the future extension and its event horizon. Noting that $t \to -t$ implies

$$t \to -t \quad \Rightarrow \quad v \to -u \quad , \quad dr/dt \to -dr/dt \ , \tag{27.113}$$

   it is clear that one will have access to this new region when working with the outgoing Eddington-Finkelstein coordinates (u, r) instead, and backtracking outgoing lightrays beyond $t = -\infty$.

   Indeed, when one uses the coordinates (u, r) instead of (v, r), the lightcones in Figure 18 are flipped (either up-down or left-right), and one can now pass through the horizon along future directed paths only in the outgoing direction of increasing r.

   The new region of space-time covered by the coordinates (u, r) is thus definitely different from the new region we uncovered using (v, r) even though both of them lie 'behind' $r = 2m$. In fact, this one is a past extension (beyond $t = -\infty$) of

the original Schwarzschild 'patch' of space-time. In this patch, the region behind $r = 2m$ acts like the opposite (time-reversal) of a black hole (a white hole) which cannot be entered on any future-directed path.

As we will discuss later, in section 29.3, this white hole region is unphysical, i.e. an artefact of the idealisation of an eternal black hole metric. The future black hole region, however, is definitely of relevance as it can be created by gravitational collapse.

2. The above statement about the behaviour of lightrays for $r = r_s = 2m$ and $r < r_s$ can be phrased somewhat more invariantly and geometrically in terms of the *expansion* (12.101), (12.102)

$$\theta_\ell = \tfrac{1}{2} s^{\alpha\beta} L_\ell s_{\alpha\beta} = \frac{1}{\sqrt{s}} L_\ell \sqrt{s} \quad . \tag{27.114}$$

of a null geodesic congruence introduced in connection with the Raychaudhuri equation in section 12.4, and measuring the change in the cross-sectional area element $\sqrt{s}$ of the congruence along the congruence.

To that end let us introduce the null vector fields

$$n = -\partial_r \quad , \quad \ell = \partial_v + \tfrac{1}{2} f(r) \partial_r \quad . \tag{27.115}$$

It is easy to check that these are 2 linearly independent future-pointing null vector-fields, n corresponding to ingoing lightrays and ℓ to (would-be) outgoing lightrays, cross-normalised to $n_\alpha \ell^\alpha = -1$,

$$n_\alpha n^\alpha = \ell_\alpha \ell^\alpha = 0 \quad , \quad n_\alpha \ell^\alpha = -1 \quad . \tag{27.116}$$

With $\sqrt{s} = r^2 \sin\theta$, one finds from (27.114) that

$$\theta_n = -\frac{2}{r} \quad , \quad \theta_\ell = \frac{r f(r)}{r^2} = \frac{r - 2m}{r^2} \quad . \tag{27.117}$$

In particular, in Minkowski space one has the standard behaviour that the ingoing radial null congruence always has negative expansion (it is contracting) while the outgoing radial null congruence has positive expansion (it is expanding),

$$f(r) = 1 \quad \Rightarrow \quad \theta_n = -\frac{2}{r} < 0 \quad , \quad \theta_\ell = +\frac{1}{r} > 0 \quad . \tag{27.118}$$

While one still has $\theta_n < 0$ in the Schwarzschild geometry (ingoing lightrays are contracting),

$$f(r) = 1 - \frac{2m}{r} \quad \Rightarrow \quad \theta_n < 0 \quad \forall\, r \quad , \tag{27.119}$$

for the congruence ℓ one has

$$f(r) = 1 - \frac{2m}{r} \quad \Rightarrow \quad \begin{cases} \theta_\ell > 0 & r > r_s \\ \theta_\ell = 0 & r = r_s \\ \theta_\ell < 0 & r < r_s \end{cases} \tag{27.120}$$

Thus $r = r_s$ is characterised by the fact that the "outgoing" lightrays have zero expansion. We will come back to this, and the related notion of *trapped surfaces*, briefly in the context of the discussion of horizons of black holes in section 32.

Since we now have at our disposal

- the defining feature of a black hole, namely the existence of an event horizon which causally seals off the interior (black hole) region from the outside (we will discuss this in more general terms in section 32.4),
- and the prime example of a black hole gravitational field, namely that described by the future extension of the Schwarzschild metric,

let us briefly, to conclude this section, discuss (and equally briefly dispose of) several popular misconceptions about black holes that are unfortunately quite common in the pop-sci and sci-fi literature:

1. black holes as cosmic vacuum cleaners

   Often a black hole is pictured as an object travelling through space and in the process violently sucking in everything around it. This is quite misleading: at a coordinate distance r in the Schwarzschild metric, the gravitational field (i.e. the metric) is the same regardless of whether in the interior there is a star of mass $M = m/G_N$ or a black hole of the same mass (there is no extra "black hole force" that enters the scene, it is all just gravity).

   In particular, if our sun, say, suddenly collapsed to form a black hole (as we will discuss in section 29, this is the way black holes are believed to form in astrophysical processes, even though this will not be the fate of our sun), then quite a number of things will change for humanity but what will not change appreciably (if the mass is appropriately conserved in this process) is the outside gravitational field and thus e.g. the orbit of the earth.

2. "locking in" of lightrays by black holes as a consequence of extremely strong gravitational fields

   It is natural to think that the properties of black holes are due to extremely strong gravitational effects at or near the horizon which prevent even lightrays from escaping the gravitational field. This point of view is of course not completely incorrect, and is occasionally reinforced by a classical "derivation" of the Schwarzschild radius as the radius at which the escape velocity from an object of mass M with that radius equals the speed of light ($v^2/2 = G_N M/r$ and $v = c$ imply $r = 2G_N M/c^2 = 2m$; I hope I do not have to tell you that you should not take this "calculation" seriously). More to the point, perhaps, we had seen that

static observers close to $r = 2m$ require very large accelerations to balance what these observers perceive to be the gravitational attraction.

Nevertheless, this is somewhat misleading since this gravitational attraction is observer-dependent (e.g. there is none for a freely falling observer) and thus does not provide us with an objective measure of the strength of the gravitational field. Objective information about the gravitational field is contained in the tidal forces, as encoded in the Riemann curvature tensor via the geodesic deviation equation discussed e.g. in section 8.4. For instance, the Kretschmann scalar (8.50) of the Schwarzschild metric is (27.163)

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \sim \frac{m^2}{r^6} \ . \tag{27.121}$$

This shows that the strength of the gravitational field in the Schwarzschild geometry, as measured by tidal forces, is $\sim m/r^3$ (as in the Newtonian theory). Near the horizon at $r = 2m$ this is

$$m/r^3 \sim 1/m^2 \tag{27.122}$$

and can thus be arbitrarily weak for sufficiently massive (and large) black holes. Related to this, we will estimate in section 29.3 the average density of an object with the size of its Schwarzschild radius and will see that for sufficiently massive objects (e.g. galaxies) this density can be as small as one likes.

The crucial point, as we will discuss in more detail in section 32, is that by definition a black hole is characterised in terms of global properties of space-time (in the sense of a region of space-time that is invisible to an asymptotic observer), and these are not necessarily unambiguously detectable by local observers. In particular, as we will see, event horizons can exist in (and emerge from) flat Minkowskian regions of space-time, in which there is definitely no trace of a strong gravitational field (at that time!).

27.6 Eddington Time Coordinate and Kerr-Schild Form of the Metric

Ordinary space-time diagrams are more familiar (and therefore more intuitive) than space-null diagrams such as the above (r, v)-diagram (in which, for example, in the asymptotically flat regime $r \to \infty$ the lightcone has slopes 0 and 2 rather than the usual $\pm 45°$ slopes ± 1). In the present case this can easily be rectified by introducing a new time-coordinate $\tilde{t}$ instead of v by the relation

$$v = t + r^* = \tilde{t} + r \ , \tag{27.123}$$

i.e.

$$\tilde{t} = t + 2m \log(r/2m - 1) \ . \tag{27.124}$$

This time coordinate is also known as the *Eddington time coordinate* (it was discovered and used first by Eddington, and then rediscovered by Finkelstein, but neither of them wrote down explicitly the null form of the metric that is now known as the Eddington-Finkelstein form of the metric and which we discussed above).

Initially the metric looks perhaps somewhat less illuminating in the $(\tilde{t}, r)$-coordinates,

$$ds^2 = -(1-2m/r)d\tilde{t}^2 + (4m/r)d\tilde{t}\,dr + (1+2m/r)dr^2 + r^2 d\Omega^2 \quad , \tag{27.125}$$

but the lightcones and the horizon now have the following simple and easy to visualise description, which follows from factorising the line element as

$$ds^2 = (d\tilde{t} + dr)(-(1-2m/r)d\tilde{t} + (1+2m/r)dr) + r^2 d\Omega^2 \quad , \tag{27.126}$$

namely:

- the ingoing side of the lightcone is described by

$$d\tilde{t} + dr = 0 \quad \Leftrightarrow \quad d\tilde{t}/dr = -1 \quad \Leftrightarrow \quad v = \text{const.} \tag{27.127}$$

  (as we already knew) and is thus at $-45°$ everywhere in a $(\tilde{t}, r)$-diagram.

- the "outgoing" side of the lightcone is described by

$$ds^2 = -(1-2m/r)d\tilde{t} + (1+2m/r)dr = 0 \quad \Leftrightarrow \quad d\tilde{t}/dr = (1+2m/r)/(1-2m/r) \quad . \tag{27.128}$$

  Therefore $d\tilde{t}/dr = +1$ for $r \to \infty$ (slope $+1$), so lightcones have the standard Minkowskian form there, $d\tilde{t}/dr \to \infty$ for $r \to 2m$, and $d\tilde{t}/dr \to -1$ for $r \to 0$.

- in particular, the horizon is (again) vertical in such a diagram, with the (would-be) outgoing side of the lightcone vertical and tangent to the horizon, while the lightcone degenerates as $r \to 0$.

Nice diagrams that you can find in many places depicting the collapse of a spherically symmetric star to a black hole and the formation of the horizon typically (either implicitly or explicitly) use these $(\tilde{t}, r)$-coordinates.

Remarks:

1. This coordinate transformation is once again of the general form $T(t, r) = t + \psi(r)$, whose effect on the Schwarzschild metric was studied in some detail in section 27.2. In particlar, the time coordinate $\tilde{t}$ (27.124) already appeared in (27.25).

2. Note that in these coordinates the metric can be split as

$$\begin{aligned} ds^2 &= -(1-2m/r)d\tilde{t}^2 + (4m/r)d\tilde{t}\,dr + (1+2m/r)dr^2 + r^2 d\Omega^2 \\ &= \left(-d\tilde{t}^2 + dr^2 + r^2 d\Omega^2\right) + \frac{2m}{r} dv(\tilde{t}, r)^2 \quad . \end{aligned} \tag{27.129}$$

Thus it has the curious property that in components it has the form

$$g_{\alpha\beta} = \eta_{\alpha\beta} + \frac{2m}{r}\partial_\alpha v \partial_\beta v \tag{27.130}$$

where $\eta_{\alpha\beta}$ is the Minkowski metric (here written in spatial spherical coordinates) and $\partial_\alpha v$ is null with respect to the inverse Minkowski metric $\eta^{\alpha\beta}$,

$$\eta^{\alpha\beta}\partial_\alpha v \partial_\beta v = \eta^{vv} = 0 \tag{27.131}$$

because $v = \tilde{t} + r$ is a null coordinate for the Minkowski metric $-d\tilde{t}^2 + dr^2 + (\ldots)$. This is the Schwarzschild metric in what is known as *Kerr-Schild form*. I will therefore also occasionally refer to $\tilde{t}$ as the Kerr-Schild time coordinate (but, as mentioned above, $\tilde{t}$ is commonly also known as the Eddington time coordinate).

3. By introducing standard inertial (Cartesian) coordinates $\tilde{x}^\alpha = (\tilde{t}, \vec{x})$ for Minkowski space, the metric can now also be written in the form

$$ds^2 = -d\tilde{t}^2 + d\vec{x}^2 + \frac{2m}{r}(\ell_\alpha d\tilde{x}^\alpha)^2 \tag{27.132}$$

where $r^2 = \vec{x}^2$, and ℓ_α has the components

$$\ell_\alpha = \partial_\alpha(\tilde{t} + r) = (1, x^k/r) \ . \tag{27.133}$$

4. There is evidently also a corresponding Kerr-Schild form of the metric in outgoing Eddington-Finkelstein coordinates, now with

$$u = t - r^* = \tilde{t} - r \quad \Rightarrow \quad \tilde{t} = t - 2m\log(r/2m - 1) \ , \tag{27.134}$$

so this is not the same $\tilde{t}$ as before, namely

$$ds^2 = \left(-d\tilde{t}^2 + dr^2 + r^2 d\Omega^2\right) + \frac{2m}{r}du(\tilde{t}, r)^2 \ . \tag{27.135}$$

More generally, metrics of the form

$$g_{\alpha\beta} = \eta_{\alpha\beta} + f(x)N_\alpha N_\beta \quad \text{with} \quad \eta^{\alpha\beta}N_\alpha N_\beta = \eta_{\alpha\beta}N^\alpha N^\beta = 0 \tag{27.136}$$

are known as *Kerr-Schild metrics*. This ansatz for the metric played an important role in the search for other exact solutions of the Einstein equations.

The form (27.136) implies that

1. $N^\alpha \equiv \eta^{\alpha\beta}N_\beta$ is also null with respect to $g_{\alpha\beta}$:

$$g_{\alpha\beta}N^\alpha N^\beta = \eta_{\alpha\beta}N^\alpha N^\beta + f(x)(N_\alpha N^\alpha)(N_\beta N^\beta) = 0 \tag{27.137}$$

2. the inverse metric is given by

$$g^{\alpha\beta} = \eta^{\alpha\beta} - f(x)N^\alpha N^\beta \tag{27.138}$$

Indeed, for $g^{\alpha\beta}$ of this form one has

$$\begin{aligned} g^{\alpha\beta}g_{\beta\gamma} &= (\eta^{\alpha\beta} - f(x)N^\alpha N^\beta)(\eta_{\beta\gamma} + f(x)N_\beta N_\gamma) \\ &= \delta^\alpha_{\ \gamma} - f(x)N^\alpha N_\gamma + f(x)N^\alpha N_\gamma = \delta^\alpha_{\ \gamma} \ . \end{aligned} \tag{27.139}$$

3. the contravariant components of N_α with respect to $g_{\alpha\beta}$ are the same as those with respect to $\eta_{\alpha\beta}$:

$$g^{\alpha\beta}N_\beta = \eta^{\alpha\beta}N_\beta - f(x)N^\alpha N^\beta N_\beta = \eta^{\alpha\beta}N_\beta \ , \tag{27.140}$$

so that the notation N^α is unambiguous.

27.7 Kruskal-Szekeres Coordinates

Using Eddington-Finkelstein coordinates, we have discovered two "new" regions of the space-time. Are there still other regions of space-time to be discovered? The answer is yes. Not only is this suggested by the analogy with Rindler and Minkowski space (but this is after all just an analogy at the moment). We had actually already seen a "doubling" of the Schwarzschild patch in the discussion of the Schwarzschild metric in isotropic coordinates (see the comment at the end of section 24.5). One way to rediscover this region from the present perspective would be to study spacelike rather than null geodesics. Alternatively, let us try to short-cut this somewhat and let us guess how one might be able to describe the maximal extension of space-time.

The first guess might be to use the coordinates u and v simultaneously, instead of r and t. In these coordinates, the metric takes the form

$$ds^2 = -(1 - 2m/r)du\ dv + r^2 d\Omega^2 \ , \tag{27.141}$$

with $r = r(u, v)$. While this is a good idea, the problem is that in these coordinates the horizon is once again infinitely far away, at $u = +\infty$ or $v = -\infty$ (i.e. at $2r^* = v - u = -\infty$). We can rectify this by introducing coordinates U and V with

$$U = -\mathrm{e}^{-u/4m} \quad , \quad V = \mathrm{e}^{v/4m} \tag{27.142}$$

say, so that the horizon is now at either $U = 0$ or $V = 0$. To better understand why we choose these coordinates and why exactly this factor in the exponent (and not any other positive number, which so far would have had the same effect of moving the horizon to $U = 0$ or $V = 0$), note that

$$\frac{v - u}{4m} = \frac{r}{2m} + \log\left(\frac{r}{2m} - 1\right) \ , \tag{27.143}$$

so that the prefactor $f(r) = 1 - 2m/r$ can be written as

$$1 - \frac{2m}{r} = \frac{2m}{r}\left(\frac{r}{2m} - 1\right) = \frac{2m}{r}\mathrm{e}^{-r/2m}\mathrm{e}^{(v-u)/4m} \quad . \tag{27.144}$$

Thus the metric can then be written as

$$ds^2 = \frac{2m}{r}\mathrm{e}^{-r/2m}\left(\mathrm{e}^{v/4m}dv\right)\left(-\mathrm{e}^{-u/4m}du\right) + r(u,v)^2 d\Omega^2 \quad . \tag{27.145}$$

It is now clear why we made the choice (27.142). The metric now takes the simple form

$$ds^2 = -\frac{32m^3}{r}\mathrm{e}^{-r/2m}dUdV + r(U,V)^2 d\Omega^2 \quad , \tag{27.146}$$

with $r = r(U,V)$ given implicitly by

$$UV = -\mathrm{e}^{(v-u)/4m} = -\mathrm{e}^{r^*/2m} = -(r/2m-1)\mathrm{e}^{r/2m} \tag{27.147}$$

and $t = t(U,V)$ explicitly by

$$U/V = -\mathrm{e}^{-(u+v)/4m} = -\mathrm{e}^{-t/2m} \quad . \tag{27.148}$$

Finally, we pass from the null coordinates (U,V) (meaning that ∂_U and ∂_V are null vectors) to more familiar timelike and spacelike coordinates (T,X) defined, in analogy with $(u,v) = t \mp r^*$, by

$$U = T - X \quad , \quad V = T + X \quad , \tag{27.149}$$

in terms of which the metric is

$$ds^2 = \frac{32m^3}{r}\mathrm{e}^{-r/2m}(-dT^2 + dX^2) + r^2 d\Omega^2 \quad . \tag{27.150}$$

Here $r = r(T,X)$ is now implicitly given by

$$X^2 - T^2 = (r/2m-1)\mathrm{e}^{r/2m} \quad . \tag{27.151}$$

Chasing through the above sequence of coordinate transformations

$$(t,r) \to (t,r^*) \to (u,v) \to (U,V) \to (T,X) \quad , \tag{27.152}$$

one finds that the coordinate transformation $(t,r) \to (T,X)$ is explicitly, and in its full glory, given by

$$\begin{aligned} X(t,r) &= \tfrac{1}{2}(V-U) = (r/2m-1)^{1/2}\mathrm{e}^{r/4m}\cosh t/4m \\ T(t,r) &= \tfrac{1}{2}(U+V) = (r/2m-1)^{1/2}\mathrm{e}^{r/4m}\sinh t/4m \quad , \end{aligned} \tag{27.153}$$

Remarks:

1. In these coordinates, the original "Schwarzschild patch" $r > 2m$, the region of validity of the Schwarzschild coordinates, corresponds to the region $-\infty < U < 0$, $0 < V < \infty$, or, in terms of X and T, to the region $X > 0$ and $X^2 - T^2 > 0$, or $|T| < X$. As Figure 19 shows, this 'Schwarzschild patch' is mapped to the first quadrant of the Kruskal-Szekeres metric, bounded by the lines $X = \pm T$.

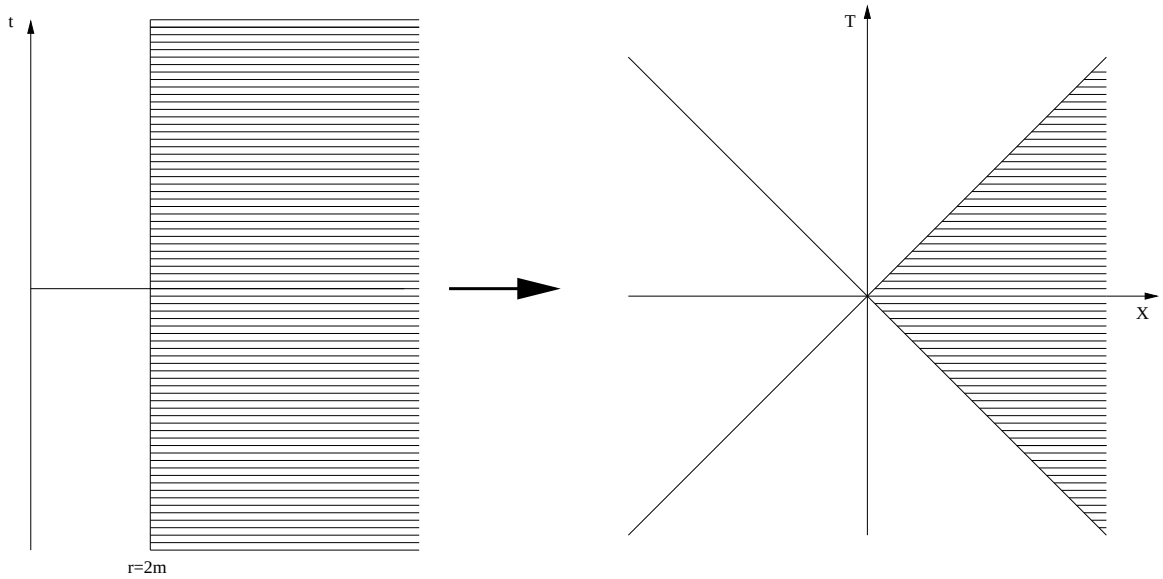


Figure 19: Schwarzschild patch in the Kruskal-Szekeres metric: the half-plane $r > 2m$ is mapped to the quadrant between the lines $X = \pm T$ in the Kruskal-Szekeres metric.

2. As in Minkowski space, null lines are given by $X = \pm T + const.$. In particular, therefore, lightcones have the standard Minkowskian form (slope ± 1) everywhere.

3. Surfaces of constant r are given by the lines (hyperboloids) $X^2 - T^2 = const.$

4. In particular, the boundary of the Schwarzschild patch is the horizon $r = 2m$. In terms of (U, V) this is mapped to $UV = 0$ which is the union of the two lines (null surfaces) $U = 0$ and $V = 0$, and in terms of (T, X) one has

$$r = 2m \quad \Rightarrow \quad X = \pm T \ , \tag{27.154}$$

5. In the region $r \gtrsim 2m$, the above coordinate transformation (27.153) reduces approximately (and up to constant factors) to the transformation (26.80) between the Rindler-like coordinates in that region and Minkowski coordinates,

$$T \sim t_M \sim (r - 2m)^{1/2} \sinh t/4m \quad , \quad X \sim x_M \sim (r - 2m)^{1/2} \cosh t/4m \tag{27.155}$$

6. In these coordinates the metric is now manifestly completely non-singular and regular not only at the horizons but everywhere except possibly at $r = 0$. We will discuss this extension of the Schwarzschild space-time in more detail in the next section.

7. Minor cosmetic improvements can be obtained by introducing, instead of U and V (27.142), the rescaled coordinates (u_K, v_K) through

$$\begin{aligned} du_K = \mathrm{e}^{-u/4m} du \quad &\Rightarrow \quad u_K = -4m\mathrm{e}^{-u/4m} = 4m\, U \\ dv_K = \mathrm{e}^{v/4m} dv \quad &\Rightarrow \quad v_K = +4m\mathrm{e}^{+v/4m} = 4m\, V \ . \end{aligned} \tag{27.156}$$

In particular, the coordinates (u_K, v_K) are now dimensionful (with dimension of length) while the coordinates (U, V) are dimensionless. In terms of (u_k, v_K) the metric takes the form

$$ds^2 = -\frac{2m}{r}\mathrm{e}^{-r/2m} du_K dv_K + r^2 d\Omega^2 \quad , \tag{27.157}$$

but then factors of $(16m^2)$ will reappear in other places. We will mostly work with the dimensionless coordinates (U, V) in the following.

8. It is also possible to further scale u_K and v_K so that the exponential prefactor has the form $\exp(-(r-2m)/2m)$ (e.g. by defining r^* with a different integration constant, $r^* \to r^* - 2m$), so that for $r \to 2m$ the metric tends to $ds^2 \to -du_k dv_k$ without additional numerical factors like e^{-1} but, as I said, this is pure cosmetics. In any case, as we will learn later on, U and V are coordinates that are naturally defined only up to affine transformations, so one choice is as good as any other.

9. Writing the implicit relation (27.147) between r and (U, V) as

$$-UV/e = (r/2m - 1)\mathrm{e}^{(r/2m - 1)} \quad , \tag{27.158}$$

one sees that r can be expressed in terms of the *Lambert function* W,

$$r(U, V) = 2m\,(W(-UV/e) + 1) \tag{27.159}$$

defined by

$$x = W(x)\mathrm{e}^{W(x)} \tag{27.160}$$

($W(x) \geq W(-1/e) = -1$, $W(0) = 0$ is the horizon).[86]

10. Note that the transformation (27.156) is strictly identical to the transformation (3.97) between Minkowski and Rindler null coordinates, with the identification $a = 1/4m$. This is much more profound than it sounds at first. In particular, we will learn in section 27.10 that $\kappa = 1/4m$ is the *surface gravity* of a black hole, i.e. a measure of the gravitational acceleration at the horizon. It is therefore this acceleration that plays the same role in the black hole context as the acceleration called a in the Rindler context. Moreover, this identity is at the heart of the deep relation between the so-called *Unruh Effect* (thermal nature of a Minkowski QFT vacuum state when seen by an accelerating observer, already mentioned in section 7.8) and the *Hawking Effect* (quantum radiation of black holes).[87]

[86] For some applications of the Lambert function in this context see e.g. K. Lake, *Some notes on the Kruskal - Szekeres completion*, `arXiv:1002.3600 [gr-qc]`, *Some further notes on the Kruskal - Szekeres completion*, `arXiv:1202.0860 [gr-qc]`.

[87] For an introduction to QFT in curved spacce-time and a discussion of these effects, see e.g. the monographs N. Birrell, P. Davies, *Quantum Fields in Curved Space*, V. Mukhanov, S. Winitzki, *Quantum Effects in Gravity*, or the on-line lecture notes J. Traschen, *An Introduction to Black Hole Evaporation*, `arXiv:gr-qc/0010055`, T. Jacobson, *Introduction to Quantum Fields in Curved Spacetime and the Hawking Effect*, `arXiv:gr-qc/0308048`, C. Krishnan, *Quantum Field Theory, Black Holes and Holography*, `arXiv:1011.5875`.

11. Further insight into the nature of Kruskal-Szekeres coordinates will be provided in section 27.10 (where we elucidate the physical significance of the coordinates (U, V)) and in section 31.9, where we outline the construction of such coordinates for more general black hole metrics.

27.8 Maximal Extension of Schwarzschild: the Kruskal Diagram

Now that we have the coordinates X and T, we can let them range over all the values for which the metric is non-singular. The only remaining singularity is at $r = 0$, which corresponds to the two sheets of the hyperboloid

$$r = 0 \quad \Leftrightarrow \quad T^2 - X^2 = 1 \quad \Leftrightarrow \quad T = \pm\sqrt{1 + X^2} \ . \tag{27.161}$$

That $r = 0$ is indeed a real singularity that cannot be removed by a coordinate transformation can be shown by calculating some invariant of the curvature tensor, like the Kretschmann scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ (8.50). On purely dimensional grounds, from

$$\left(\partial_r^2(1 - 2m/r)\right)^2 \sim m^2/r^6 \ , \tag{27.162}$$

say, one would expect K to be proportional to m^2/r^6, the crucial feature being that the constant of proportionality is not zero, explicit calculations (this is a doable but thoroughly unenlightning exercise) showing that

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48\frac{m^2}{r^6} \ . \tag{27.163}$$

Thus the geometry is genuinely singular at $r = 0$. Nevertheless, since the metric is non-singular for all values of (X, T) subject to the constraint $r > 0$ or $T^2 - X^2 < 1$, there is no physical reason to exclude the regions in the other quadrants also satisfying this condition.

By including them, we obtain the *Kruskal-Szekeres Extension* of the Schwarzschild metric, displayed in the *Kruskal diagram* in Figure 20. This extension of the Schwarzschild metric was discovered independently by Kruskal and Szekeres in 1960 and presents us with an amazingly rich and complex picture of what originally appeared to be a rather simple (and perhaps even dull) solution to the Einstein equations. It can be shown that this represents the maximal analytic extension of the Schwarzschild metric in the sense that every affinely parametrised geodesic can either be continued to infinite values of its parameter or runs into the singularity at $r = 0$ at some finite value of the affine parameter.[88]

In addition to the Schwarzschild patch, quadrant I, we have three other regions, living in the quadrants II, III, and IV, each of them having its own peculiarities. Note that

[88] One can also show that any spherically symmetric solution of the vacuum Einstein equations is locally isometric to some domain of the Schwarzschild-Kruskal solution. For a proof of this variant of a generalised Birkhoff theorem see e.g. N. Straumann, *General Relativity*.

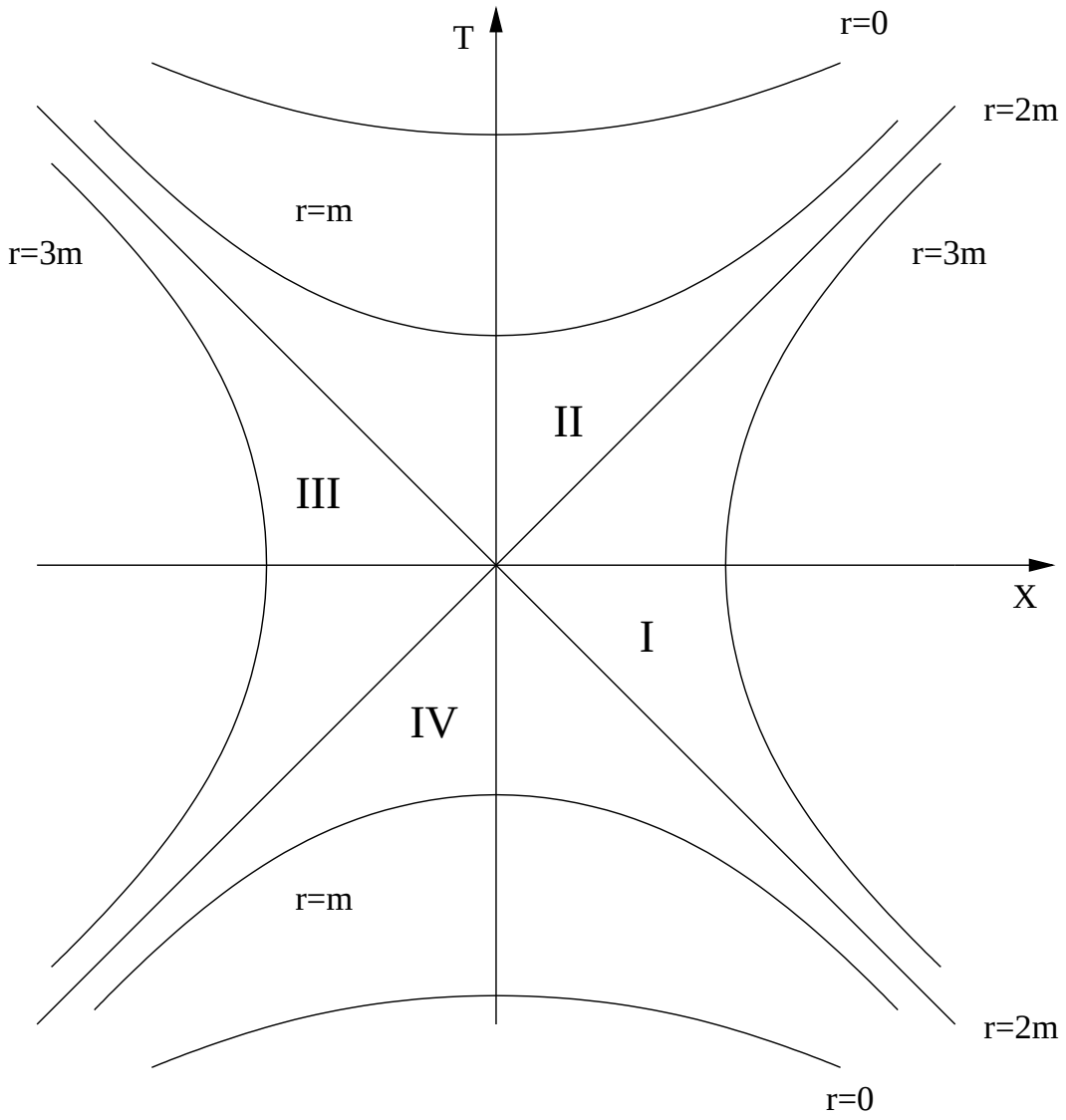


Figure 20: Complete Kruskal-Szekeres universe. Diagonal lines are null, lines of constant r are hyperbolas. Region I is the Schwarzschild patch, separated by the horizon from regions II and IV. The Eddington-Finkelstein coordinates (v, r) cover regions I and II, (u, r) cover regions I and IV. Regions I and III are filled with lines of constant $r > 2m$. They are causally disconnected. Observers in regions I and III can receive signals from region IV and send signals to region II. An observer in region IV can send signals into both regions I and III (and therefore also to region II) and must have emerged from the singularity at $r = 0$ at a finite proper time in the past. Any observer entering region II will be able to receive signals from regions I and III (and therefore also from IV) and will reach the singularity at $r = 0$ in finite time. Events occuring in region II cannot be observed in any of the other regions.

obviously the conversion formulae from $(r,t) \to (X,T)$ in the quadrants II, III and IV differ from those given above for quadrant I. E.g. in region II one can use Schwarzschild(-like) coordinates in which the metric reads

$$ds^2 = \left(\frac{2m}{r} - 1\right) dt^2 - \left(\frac{2m}{r} - 1\right)^{-1} dr^2 + r^2 d\Omega^2 \tag{27.164}$$

(these are not the same coordinates as those in patch I, as we have seen we cannot continue the Schwarzschild coordinates across the horizon), and in this quadrant (where r is a time coordinate etc.) the relation between Schwarzschild and Kruskal coordinates is

$$\begin{aligned} X &= (1 - r/2m)^{1/2} \mathrm{e}^{r/4m} \sinh t/4m \\ T &= (1 - r/2m)^{1/2} \mathrm{e}^{r/4m} \cosh t/4m \ . \end{aligned} \tag{27.165}$$

To get acquainted with the Kruskal diagram, let us note the following basic facts (some of which we had already noted for region I in the previous section).

1. Null lines are diagonals $X = \pm T + const.$, just as in Minkowski space. This greatly facilitates the exploration of the causal structure of the Kruskal-Szekeres metric.

2. In particular, the horizon corresponds to the two lines $X = \pm T$.

3. Lines of constant r are hyperbolas. For $r > 2m$ they fill the quadrants I and III, for $r < 2m$ the other regions II and IV.

4. In particular, the singularity at $r = 0$ is given by the two sheets of the hyperbola $T^2 - X^2 = 1$.

5. Notice in particular also that in regions II and IV worldlines with $r = const.$ are no longer timelike but spacelike. Including the transverse 2-sphere, this should be rephrased as either "lines of constant (r, θ, ϕ) are spacelike for $r < 2m$" or "the surfaces of constant r are spacelike for $r < 2m$".

6. Lines of constant Schwarzschild time t are straight lines through the origin. E.g. in region I one has $X = (\coth(t/4m))T$, with the future horizon $X = T$ corresponding, as expected, to $t \to \infty$.

7. The Eddington-Finkelstein coordinates (v, r) cover the regions I and II, the coordinates (u, r) the regions I and IV.

8. Quadrant III is completely new and is separated from region I by a spacelike distance. That is, regions I and III are causally disconnected. This is the region that already prematurely made a brief appearance when we analysed the coordinate transformation between Schwarzschild and isotropic coordinates at the end of section 24.5. Thus isotropic coordinates actually cover the regions I and III.

Now let us see what all this tells us about the physics of the Kruskal-Szekeres metric.

- An observer in region I (the familiar patch) can send signals into region II and receive signals from region IV.
- The same is true for an observer in the causally disconnected region III.
- Once an observer enters region II from, say, region I, he cannot escape from it anymore and he will run into the catastrophic region $r = 0$ in finite proper time.
- As a reward for his or her foolishness, between having crossed the horizon and being crushed to death, our observer will for the first time be able to receive signals and meet observers emerging from the mirror world in region III.
- Events occurring in region II cannot be observed anywhere outside that region (black hole).
- Finally, an observer in region IV must have emerged from the (past) singularity at $r = 0$ a finite proper time in the past and can send signals and enter into either of the regions I or III.

An even better visualisation of the causal *and global* structure of the maximally extended Schwarzschild solution is provided by its *Carter-Penrose Conformal Diagram* (or *Penrose diagram* for short), given in Figure 21. Its construction (and the notation used here) will be explained in section 28.

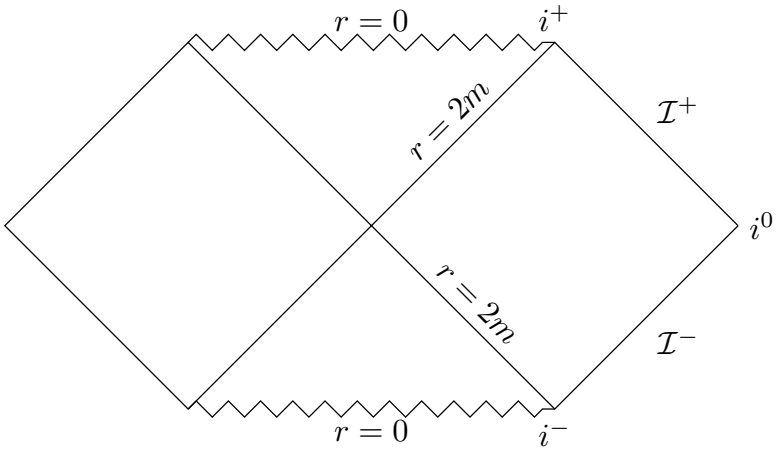


Figure 21: Carter-Penrose Conformal Diagram of the maximal Kruskal-Szekeres extension of the Schwarzschild space-time. For more details, see section 28.

All in all, the picture that we have uncovered of the complete maximally extended Schwarzschild space-time is quite intricate and rich, but also somewhat peculiar (to say the least). In particular, the mirror region III and the white hole region IV appear to

be quite unphysical. As we will see in section 29.3, reassuringly the existence of these regions is an artefact of an eternal black hole solution and these regions do not actually exist for astrophysical black hole solutions that are formed by gravitational collapse.

27.9 Properties of the Asymptotically Timelike Killing Vector

An interesting aspect of the Kruskal-Szekeres geometry is its dynamical character. This may appear to be a strange thing to say since we explicitly started off with a static metric, but this statement applies only to region I (and its mirror III). An investigation of the behaviour of spacelike slices analogous to that we performed for region I (see Figure 11) reveals a dynamical picture of continuing gravitational collapse in region II. In simple terms, the loss of staticity can be understood by noting that the timelike Killing vector field $\xi = \partial_t$ of region I, when expressed in terms of Kruskal coordinates, becomes null on the horizon and spacelike in region II.

Indeed it is easy to check from (27.147) and (27.148) that the time-translation symmetry $(t, r) \to (t + c, r)$ of the Schwarzschild patch corresponds to the transformation

$$U \to \mathrm{e}^{-c/4m} U \qquad V \to \mathrm{e}^{c/4m} V \ . \tag{27.166}$$

This is a boost in the (U, V) or (T, X)-plane which leaves the entire Kruskal metric invariant since

$$dU dV = dT^2 - dX^2 \tag{27.167}$$

is invariant and $r = r(U, V)$ depends on U and V only via the boost-invariant (time-independent) quantity UV (27.147),

$$UV = T^2 - X^2 = -(r/2m - 1)\mathrm{e}^{r/2m} \quad \Rightarrow \quad r = r(T^2 - X^2) \ . \tag{27.168}$$

Thus this symmetry is generated by the Killing vector

$$\xi = (V\partial_V - U\partial_U)/4m = (X\partial_T + T\partial_X)/4m \ . \tag{27.169}$$

It follows that ξ has norm proportional to $T^2 - X^2$,

$$||\xi||^2 = \frac{2m}{r}\mathrm{e}^{-r/2m}(T^2 - X^2) \ . \tag{27.170}$$

There are thus three different cases to consider, each of them interesting in its own right:

- ξ timelike

  ξ is timelike in the original region I (and in the mirror region III), with (27.151)

  $$I: \quad ||K||^2 = -\frac{2m}{r}\mathrm{e}^{-r/2m}(\frac{r}{2m} - 1)\mathrm{e}^{r/2m} = -(1 - \frac{2m}{r}) \ , \tag{27.171}$$

  confirming that $\xi = \partial_t$ in region I. We thus recover the statement that the Schwarzschild metric in the Schwarzschild patch is static.

- ξ spacelike

  ξ is spacelike in region II. Thus region II has no timelike Killing vector field, therefore cannot possibly be static, but has instead an additional spacelike Killing vector field - cf. the Remark 1 in section 24.6 in connection with Birkhoff's theorem.

  Related to this is the fact, already mentioned above, that in regions II and IV the slices of constant r are no longer timelike but spacelike surfaces. Thus they are analogous to, say, constant t or T slices for $r > 2m$. Just as it does not make sense to ask "where is the slice $t = 1$?" (say), only "when is $t = 1$?", or "where is $r = 3m$?", in these regions it makes no sense to ask "where is $r = m$?", only "when is $r = m$?".

- ξ null

  ξ is null on the horizon. This turns out to be the most interesting case, and turns out to be one of the characteristic features of static black holes in general, and therefore I will elaborate on this a bit in the following.

27.10 First Encounter with Killing Horizons and Surface Gravity

In this section, we will perform a rather pedestrian analysis of some properties of the horizon which are related to the fact that the Killing vector ξ becomes null on this null surface. These properties can be derived and understood without a general knowledge of the geometry of null hypersurfaces, as discussed in section 17, but it is useful to keep in mind that they are just special cases of the properties of null hypersurfaces, and of more general Killing horizons (to be discussed later on in section 32.5).

In particular, from section 17.2 one knows on general grounds that the integral curves of the normal vector ξ are (possibly non-affinely parametrised) null geodesics, but it is instructive to rederive this here from scratch, and from a slightly different perspective.

ξ is null on (and tangent to) the horizon $T = \pm X$ or $UV = 0$. In fact it reduces to $\xi = (V/4m)\partial_V = \partial_v$ on the horizon $U = 0$ (and to $\xi = \partial_u$ on $V = 0$). As a consequence, due to the characteristic peculiarities of null hypersurfaces (section 17), ξ is also normal to the horizon. In this context (or from this perspective) the horizon is known as a *Killing horizon* (the locus where a Killing vector becomes null and orthogonal to a null surface).

Therefore ξ generates translations along the horizon, $v \to v + c$, and is called the *null generator* of (this branch of) the horizon. Thus v can naturally be used as a coordinate there.

Evidently, ξ vanishes at the "point" $U = V = 0$ where the other horizon $V = 0$ branches off. Actually this is of course not a point but a perfectly respectable 2-sphere

of radius $r = 2m$ known as the *bifurcation surface* of the Killing horizon of the Schwarzschild geometry, and ξ vanishing means that this 2-sphere is invariant under the time-translation generated by ξ, something that is also evident from (27.166). On $U = 0$ this branching point lies at $V \to 0 \Rightarrow v \to -\infty$, so the Eddington-Finkelstein coordinate $v \in (-\infty, +\infty)$ only covers the half-line $U = 0, V > 0$ of the horizon.

On the other hand the line $U = 0$ is itself an "outgoing" (radial) null geodesic, but in light of the above v cannot possibly be an affine parameter along that geodesic (the affine parameter should not reach infinite values half-way along the geodesic). The failure of v to be an affine parameter on the horizon can be quantified by calculating the acceleration $\nabla_\xi \xi = \nabla_{\partial t} \partial_t$ (5.98) of the (integral curves of the) Killing vector and then taking the limit $r \to 2m$. This calculation is quite painless in Eddington-Finkelstein coordinates (v, r) in which $\xi = \partial_v$ everywhere,

$$\xi = \partial_t = (\partial_t v)\partial_v + (\partial_t r)\partial_r = \partial_v \tag{27.172}$$

($\xi = \partial_v$ is evidently a Killing vector because the components of the metric do not depend on v in Eddington-Finkelstein coordinates). Then the acceleration is

$$\nabla_\xi \xi \equiv (\nabla_\xi \xi)^\alpha \partial_\alpha = \Gamma^\alpha_{vv} \partial_\alpha = (f'/2)(f\partial_r + \partial_v) \tag{27.173}$$

where $f(r) = 1 - 2m/r$. Thus for $r \to 2m$ one finds

$$\lim_{r\to 2m} \nabla_\xi \xi = \left(\tfrac{1}{2} f'(r)|_{r=2m}\right) \xi = \left(\frac{m}{r^2}|_{r=2m}\right) \xi = \tfrac{1}{4m} \xi \ . \tag{27.174}$$

Remarks:

1. Since we have $\nabla_\xi \xi \sim \xi$ on the horizon, this shows first of all that there it generates a non-affinely parametrised geodesic (see (2.136) and (5.100)). Moreover, we see that one interpretation of the ubiquitous factor

$$\kappa = \frac{1}{4m} \tag{27.175}$$

is that it measures the *inaffinity*, i.e. the failure of v to be an affine parameter on the horizon,

$$\lim_{r\to 2m} \nabla_\xi \xi = \kappa \xi \ . \tag{27.176}$$

2. Another interpretation brought out by the same calculation (27.174) is that $\kappa = 1/4m$ is the *surface gravity* which measures the strength of the gravitational force (acceleration) $\mathsf{a}(r)$ (26.11) acting on a static observer at the horizon, but as measured at infinity (by taking into account the redshift factor $f(r)^{1/2}$),

$$\kappa = \lim_{r\to 2m} f(r)^{1/2} \mathsf{a}(r) = \tfrac{1}{2} f'(r)|_{r=2m} = 1/4m \ . \tag{27.177}$$

Anyway, to return to the beginning of this story, we have seen that v is not an affine parameter along the horizon. It turns out, however, that the Kruskal coordinate V is an affine parameter there (and this is one way of understanding why Kruskal coordinates are so natural for exploring the causal structure of the metric), meaning that the null curve

$$x^\alpha(\lambda) = (U(\lambda), V(\lambda), \theta(\lambda), \phi(\lambda)) = (0, \lambda, \theta_0, \phi_0) \tag{27.178}$$

is an affinely parametrised null geodesic. This follows on the nose from (27.174) and the result (2.138) of section 2.7. Noting that κ is constant (actually not just along the geodesic but on the entire horizon, but the former is all we need), (2.138) with $\tau \to \lambda$ and $\sigma \to v$ becomes (dropping integration constants)

$$\frac{d\lambda}{dv} \sim \mathrm{e}^{\kappa v} \quad \to \quad \lambda(v) \sim \kappa^{-1}\mathrm{e}^{\kappa v} \sim V \ . \tag{27.179}$$

Another way to see this, which provides some more insight, is to analyse the geodesic Lagrangian and the conserved quantity associated to ξ in ingoing Eddington-Finkelstein coordinates (one could also work in Kruskal coordinates, but nothing is gained by that). The elementary steps in the calculation are the following:

- From the metric (27.82) we deduce that for a radial null-geodesics one has

  $$-f(r)\dot{v}^2 + 2\dot{v}\dot{r} = 0 \tag{27.180}$$

  where a dot denotes a derivative with respect to the affine parameter λ. The geodesics with $v = \text{const}$ describe ingoing null-geodesics and we are not interested in these, so we have

  $$-f(r)\dot{v} + 2\dot{r} = 0 \ , \tag{27.181}$$

  and since $u = t - r^* = v - 2r^*$ and $dr^*/dr = f(r)^{-1}$, this is equivalent to $\dot{u} = 0$ and, as anticipated, describes outgoing geodesics.

- The conserved quantity E associated to the Killing vector $\xi = \partial_v$ is

  $$f(r)\dot{v} - \dot{r} = E \ . \tag{27.182}$$

  From the two preceding equations we immediately deduce (with a convenient parametrisation of the integration constant)

  $$\dot{r} = E \quad \Rightarrow \quad r(\lambda) = E(\lambda - \lambda_0) + r_s \ . \tag{27.183}$$

  For the null geodesic along the horizon we are ultimately interested in, we have $r(\lambda) = r_s$, i.e. $E = 0$, but we need to approach this with some care, so we keep the general solution for now. Analogously, for v we find the equation

  $$f(r)\dot{v} = 2E \quad \Rightarrow \quad \dot{v} = \frac{2Er(\lambda)}{r(\lambda) - r_s} = 2E + \frac{2r_s}{\lambda - \lambda_0} \ . \tag{27.184}$$

We can now take the limit $E \to 0$ with impunity, and are left with

$$\dot{v} = \frac{2r_s}{\lambda - \lambda_0} \quad \Rightarrow \quad v(\lambda) = 2r_s \ln(\lambda - \lambda_0) + \text{const.} \ . \tag{27.185}$$

- The prefactor $2r_s = 4m$ is now precisely such that it cancels the factor $1/4m$ in the definition of the Kruskal coordinate V (27.142), so that

$$V(\lambda) = \mathrm{e}^{v(\lambda)/4m} = a(\lambda - \lambda_0) \ , \tag{27.186}$$

  which is precisely the statement that V is related to λ by an affine transformation. Thus V is an affine parameter along the horizon $U = 0$, as claimed.

- We also see explicitly that for other outgoing null geodesics, i.e. those with $E \neq 0$, and with solution

$$v(\lambda) = 2E\lambda + 4m \ln(\lambda - \lambda_0) + \text{const.} \ , \tag{27.187}$$

  neither v nor V is an affine parameter. However, it is apparent from the explicit solution $r(\lambda) = E\lambda + b$ given above that, for $E \neq 0$, $r(\lambda)$ is related to λ by an affine transformation, and hence for these geodesics r is an affine parameter.

Remarks:

1. We now have two natural coordinates on the future horizon $U = 0, V > 0$ which we can for instance use to measure the frequency of incoming waves. $\xi = \partial_t = \partial_v$ measures what is commonly called Killing frequency (this requires no further explanation since it is associated to the Killing vector which generates time-translations in the Schwarzschild patch), and is the natural notion of frequency to be used by static observers with 4-velocvity $u^\alpha \sim \xi^\alpha$.

   ∂_V, on the other hand, measures the so-called free fall frequency since, more or less by construction, a freely falling observer in the Schwarzschild geometry near $r = 2m$ will see approximately the Minkowski space-time metric $ds^2 \sim -dUdV$ (recall that the transformation (3.97) between Rindler and Minkowski coordinates is strictly analogous to the transformation (27.156) between Schwarzschild and Kruskal coordinates). The exponential relation (27.142) between them reflects the exponential blue- or redshift we first encountered in section 26.5.

2. As a concluding remark to this section, I cannot resist mentioning that the notion of surface gravity κ plays a crucial role in the analysis of the classical dynamics of general black holes, and even more so in the semi-classical context, since it is directly proportional to the temperature of the famous Hawking radiation of an evaporating black hole,

$$T_H = \frac{\kappa}{2\pi} = \frac{1}{8\pi G_N M} \ . \tag{27.188}$$

For more details on this and other related advanced topics I am not able to cover or do justice to here, I refer you to the references given in footnote 87 of section 27.7, as well as to the superb Cambridge lecture notes on Black Holes.[89]

27.11 From Eddington-Finkelstein to Israel(-Klösch-Strobl) Coordinates

There is one more coordinate system for the Schwarzschild geometry that I want to mention because it is quite remarkable and, equally remarkably, apparently not widely known or used. It was discovered by W. Israel in 1966, and rediscovered several times since, most recently by T. Klösch and T. Strobl in a different particularly insightful way.[90] I will introduce these coordinates in a way that is complementary to those in these articles (and that is also motivated by certain generalisations, that I will however not discuss here).

Recall that the Kruskal coordinates were based on suitably combining the outgoing and ingoing Eddington-Finkelstein coordinates. Now, more generally one frequently encounters the situation that one knows a solution either in coordinates adapted to ingoing null geodesics, or in coordinates adapted to outgoing null geodesics, but usually not both (and given one constructing the other is usually a hard problem that may have no simple analytical solution).

Thus, let us assume that we are given (or have found) the Schwarzschild metric in outgoing Eddington-Finkelstein coordinates,

$$ds^2 = -f(r)du^2 - 2du\,dr + r^2 d\Omega^2 \quad , \quad f(r) = 1 - 2m/r \ . \tag{27.189}$$

We are happy and proud of this, but we quickly realise that this cannot be the end of the story because the above coordinates provide an incomplete covering of the space-time. The simplest way to discover this is to study radial lightrays or null geodesics, governed by the two equations

$$f(r)\dot{u}^2 + 2\dot{u}\dot{r} = 0 \tag{27.190}$$

(the null condition), and

$$f(r)\dot{u} + \dot{r} = E \tag{27.191}$$

(due to the u-translation invariance). One set of null geodesics is simply given by $\dot{u} = 0$, i.e. $u = u_0$ constant, and for these one has

$$u(\tau) = u_0 \quad \Rightarrow \quad \dot{r} = E \quad \Rightarrow \quad r(\tau) = E\tau + r_0 \ . \tag{27.192}$$

[89] P. Townsend, *Black Holes*, `arXiv:gr-qc/9707012v1`.

[90] W. Israel, *New Interpretation of the extended Schwarzschild Manifold*, Phys. Rev. 143 (1966) 1016-1021; T. Klösch, T. Strobl, *Explicit Global Coordinates for Schwarzschild and Reissner-Nordstroem*, `arXiv:gr-qc/9507011`. It is also mentioned in section 15.4.4 of the directory of exact solutions, H. Stephani, D. Kramer, M. MacCallum, C. Hoenslaers, E. Herlt, *Exact Solutions to Einstein's Field Equations* (2nd Edition). See also K. Lake, *An explicit global covering of the Schwarzschild-Tangherlini black holes*, `arXiv:gr-qc/0306073`, and *Maximally extended, explicit and regular coverings of the Schwarzschild - de Sitter vacua in arbitrary dimension*, `arXiv:gr-qc/0507031` for some generalisations.

For future-oriented null geodesics one needs $E > 0$, and therefore one has $\dot{r} > 0$. These are the outgoing null geodesics to which the outgoing Eddington-Finkelstein coordinate system is adapted. Here r_0 is an integration constant which, by an affine transformation (actually just a shift) of τ, we could e.g. without loss of generality choose to be $r_0 = 2m$. Then the solution describes the outgoing null geodesics that emerge from the past event horizon at $r = 2m$ for $\tau = 0$.

The other set of (thus ingoing) null geodesics has $\dot{u} \neq 0$ and is therefore governed by the equations

$$f(r)\dot{u} + 2\dot{r} = 0 \quad \text{and} \quad f(r)\dot{u} + \dot{r} = E \ . \tag{27.193}$$

Subtracting the two one finds $\dot{r} = -E$, and therefore (again with a convenient choice of integration constant or origin of τ)

$$r(\tau) = 2m - E\tau \quad , \quad u(\tau) = -4m \log|\tau| + 2E\tau + c \qquad (-\infty < \tau < 0) \tag{27.194}$$

The restriction on τ arises because for $\tau \to 0_-$ one has

$$\tau \to 0_- \quad \Rightarrow \quad r \to 2m \quad , \quad u \to \infty \ . \tag{27.195}$$

Thus we discover that we reach $u = +\infty$ in finite (affine) time, we run out of coordinate space as the ingoing null geodesics approach $r = 2m$ (and the coordinate u is evidently not suitable for describing what happens beyond $r = 2m, u = \infty$).

Note that this locus is most certainly not the past event horizon ($r = 2m$, $u = u_0$ finite), as we know that we can only cross that horizon along future-directed paths from smaller to larger values of r (the white hole). Thus we have discovered a new barrier (which also turns out to be a horizon, and which, with hindsight, we know is the future event horizon), and we realise that the outgoing Eddington-Finkelstein coordinates do not cover the whole space-time.

Now let us assume that we do not have the luxury of being able to appeal to ingoing Eddington-Finkelstein coordinates to construct a future extension (and subsequently the maximal Kruskal-Szekeres extension) of this space-time. How could we proceed?

Since τ is finite as $u = +\infty$ is reached, it is natural to introduce something like τ as a new coordinate. Looking at the leading term of $u(\tau)$ for $\tau \to 0$, one is then led to introduce a new coordinate x through

$$u(\tau) \approx -4m \log|\tau| \quad \Rightarrow \quad u(x) = -4m \log|x| \ , \tag{27.196}$$

so that the complete range of u, $u \in (-\infty, +\infty)$, is covered as x runs over the interval $x \in (-\infty, 0)$. This clearly now permits to continue the ingoing null geodesics beyond $u = +\infty$. However, if one just replaces $u \to x$, the metric appears to be singular at $x = 0$. This can be rectified by introducing a new coordinate y through the relation

$$r - 2m = xy \ , \tag{27.197}$$

its range subject to the condition $r > 0$. To better understand the rationale for this change of variables, note that as $u \to \infty$ one has $r - 2m \sim \tau \sim x$ so that $(r-2m)/x$ remains finite in the limit - and can therefore be used as a new coordinate, the one that we have called y.

It is straightforward to see that in these coordinates the metric takes the form

$$ds^2 = \frac{8my^2}{xy+2m}dx^2 + 8m\,dx\,dy + (xy+2m)^2 d\Omega^2 \quad . \tag{27.198}$$

This is the Schwarzschild metric in *Israel coordinates*. Before discussing some of it most important properties, note that by the simple scaling $u_I = 4mx$, so that

$$u_I = 4mx = -4me^{-u/4m} \equiv u_K \tag{27.199}$$

is just the "canonically normalised" Kruskal coordinate introduced in (27.156), and the relabelling $y \to v_I$, one can put the metric into the somewhat more common "canoncially normalised" form

$$ds^2 = \frac{v_I^2}{2m((u_I/4m)v_I + 2m)}du_I^2 + 2\,du_I\,dv_I + ((u_I/4m)v_I + 2m)^2 d\Omega^2 \quad . \tag{27.200}$$

Here are some of the key-properties of this metric (to describe these I will continue to use the coordinates (x, y), i.e. the form (27.198) of the metric):

1. First of all, one sees that the metric is completely non-singular as long as $xy > -2m$, and one can therefore let x and y run over all the values for which this condition is satisfied.

2. The horizon is given by the degenerate hyperboloid

$$r = 2m \quad \Leftrightarrow \quad xy = 0: \quad \{x = 0\} \cup \{y = 0\} \tag{27.201}$$

3. Then one sees immediately that this space-time covers four distinct "patches":

   - For $x < 0$ and $y < 0$ one has $r > 2m$: this is the original Schwarzschild patch.
   - There are two regions for which x and y have opposite sign, subject to the condition $xy > -2m$: these are the white hole region behind the past horizon $(x < 0, y > 0)$, covered by the original outgoing Eddington-Finkelstein coordinates, and the new region beyond the future event horizon with $x > 0, y < 0$.
   - Finally, for $x > 0$ and $y > 0$ one also has $r > 2m$: this is a distinct region isometric to the Schwarzschild patch, the mirror region.

4. Thus the Israel coordinates provide not only a future extension of the Schwarzschild metric in Schwarzschild or outgoing Eddington-Finkelstein coordinates but actually a complete covering of the maximal Kruskal-Szekeres extension of the Schwarzschild space-time.

5. The usual Kruskal-Szekeres coordinates (U, V) are related to the Israel coordinates by

$$U \sim x \quad , \quad V \sim y\, \mathrm{e}^{xy/2m+1} \quad . \tag{27.202}$$

6. The main difference to (and advantage compared with) Kruskal-Szekeres coordinates is that for the Israel coordinates the coordinate transformation to Schwarzschild coordinates and its inverse are completely explicit (whereas the radial coordinate r is only given implicitly in terms of the Kruskal-Szekeres coordinates). E.g. for $x < 0$ one has

$$\left.\begin{array}{ll} u & = -4m\log(-x) \\ r & = xy + 2m \end{array}\right\} \quad \Leftrightarrow \quad \left\{\begin{array}{ll} x & = -\mathrm{e}^{-u/4m} \\ y & = -(r-2m)\mathrm{e}^{u/4m} \end{array}\right. \tag{27.203}$$

7. By undoing the coordinate transformation $u = t - r^*$ from Schwarzschild to Eddington-Finkelstein coordinates, one can also explicitly write the transformation from Schwarzschild coordinates (t, r) to the Israel coordinates,

$$r(x,y) = xy + 2m \quad , \quad t(x,y) = xy + 2m + \log|y/2mx| \quad . \tag{27.204}$$

8. By inspection, the metric depends only on products like xy, $dxdy$ etc. Thus it has the isometry

$$(x, y) \to (\lambda x, \lambda^{-1} y) \tag{27.205}$$

which corresponds (and reduces) to the time-translation symmetry in the Schwarzschild patch, generated by

$$\partial_t = (y\partial_y - x\partial_x)/4m \quad . \tag{27.206}$$

9. Radial null geodesics are thus goverened by the two equations

$$\frac{y^2}{xy+2m}\dot{x}^2 + \dot{x}\dot{y} = 0 \tag{27.207}$$

(the null condition), and

$$\frac{2y^2}{xy+2m}x\dot{x} + \dot{y}x - \dot{x}y = c \quad . \tag{27.208}$$

(from time-translation invariance). They are therefore of two kinds:

- One has the curves with x (as well as, of course, (θ, ϕ)) constant, so that these are straight lines in an (x, y)-diagram. For these curves, (27.208) implies that $\dot{y}$ is constant, so that y an affine parameter along these null geodesics.
- The other null geodesics are given by the relation

$$\frac{y^2}{xy+2m}\dot{x} + \dot{y} = 0 \quad . \tag{27.209}$$

Multiplying by x and subtracting twice that from (27.208), one finds

$$\dot{x}y + x\dot{y} = c \quad \Rightarrow \quad x(\tau)y(\tau) = c\tau + d \ . \tag{27.210}$$

Using this to eliminate x and $\dot{x}$ from (27.209), the equation of motion for y reduces to

$$\dot{y} = -(c/2m)y \quad \Rightarrow \quad y(\tau) = y_0 \mathrm{e}^{-(c/2m)\tau} \ , \tag{27.211}$$

or $c\tau = -2m \log(y(\tau)/y_0)$. Substituting this in (27.210), one finds that these radial null geodesics are given by the curves

$$xy = -2m \log(Cy) \tag{27.212}$$

in the (x, y)-plane, with C an arbitrary constant.

Armed with this information, it is straightforward to draw an Israel analogue of the Kruskal diagram, containing an equivalent amount of information. All in all, Israel coordinates are an attractive and easy to construct and understand alternative to Kruskal-Szekeres coordinates.

27.12 Appendix: Summary of Schwarzschild Coordinate Systems

Here is, to wrap up this section, a list of the coordinate systems that we have used to gain insight into the properties of the Schwarzschild metric. [As region III is isometric to region I, this doubling of possibilities (e.g. Schwarzschild coordinates cover I *or* III, etc.) has not been indicated in the last column. The perfectly valid but somewhat un-insightful option to use Schwarzschild coordinates only in region II, say, and variants thereof, have also not been indicated in the last column.]

Name	Line Element	Adapted to	Covers
Schwarzschild	$-f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2$	Stationary Observers	I
Tortoise	$f(r)(-dt^2 + (dr^*)^2) + r(r^*)^2 d\Omega^2$	In & Out Null Geodesics	I
Isotropic	$-(g_-(\rho)/g_+(\rho))dt^2 + g_+(\rho)^2(d\rho^2 + \rho^2 d\Omega^2)$	Stationary Observers	I,III
Painlevé - Gullstrand	$-f(r)d\tau^2 \pm 2\sqrt{\frac{2m}{r}}dr d\tau + (dr^2 + r^2 d\Omega^2)$	In/Out Timelike Geodesics	I,II / I,IV
Lemaître	$-d\tau^2 + (2m/r(\tau,\rho))d\rho^2 + r(\tau,\rho)^2 d\Omega^2$	In Timelike Geodesics	I,II
Novikov	$-d\tau^2 + \frac{R^2+1}{R^2}(r')^2 dR^2 + r(\tau,R)^2 d\Omega^2$	In Timelike Geodesics	I-IV
Eddington - Finkelstein	$-f(r)d(v/u)^2 \pm 2d(v/u)dr + r^2 d\Omega^2$	In/Out Null Geodesics	I,II / I,IV
Kerr-Schild	$-d\tilde{t}^2 + dr^2 + r^2 d\Omega^2 + \frac{2m}{r}(d(\tilde{t} \pm r))^2$	In/Out Null Geodesics	I,II / I,IV
Kruskal - Szekeres	$F(r(T,X))(-dT^2 + dX^2) + r(T,X)^2 d\Omega^2$	In & Out Null Geodesics	I-IV
Israel	$F_I(x,y)dx^2 + 8m\, dxdy + (xy+2m)^2 d\Omega^2$	In/Out Null Geodesics	I-IV

Abbreviations:

$$
\begin{aligned}
f(r) = 1 - 2m/r \quad &, \quad g_\pm(\rho) = (1 \pm m/2\rho)^2 \\
F(r) = 32m^3 r^{-1} \mathrm{e}^{-r/2m} \quad &, \quad r(\tau,\rho) = [3\sqrt{2m}(\rho - \tau)/2]^{2/3} \\
F_I(x,y) = 8my^2/(xy+2m) \quad &, \quad r(r^*) \text{ from } r^* = r + 2m\ln(r/2m - 1) \qquad (27.213) \\
r(\tau,R) \quad \text{and} \quad r' = \partial r/\partial R \quad &\text{from} \quad (27.59) \\
r(T,X) \quad &\text{from} \quad X^2 - T^2 = (r/2m - 1)\mathrm{e}^{r/2m}
\end{aligned}
$$